# Preliminary Design Report

## A Multi-Agent Robotic System for Autonomous Ice-Shell Characterization and Cryobot Site Selection on Europa

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# Contents

# 1 Executive Summary

This mission deploys a coordinated multi-agent robotic system to Europa, Jupiter's icy moon, to autonomously characterize ice-shell thickness along a surface traverse and identify an optimal cryobot insertion site, operating without real-time Earth support [1]. The system consists of a surface rover carrying a ground-penetrating radar sounder, supported by two orbital relay agents that provide terrain reconnaissance, communications relay, and cooperative localization [3]. This mission directly addresses three converging research gaps: autonomous operations in GPS-denied, high-radiation environments; multi-agent coordination and interoperability for cooperative task planning [2]; and fully autonomous approach and proximity operations without Earth-based navigational support [3].

Europa was chosen as the destination because it holds strong indicators of conditions favorable to life and contains the chemical building blocks necessary to support it [1]. Its subsurface ocean makes it one of the most promising locations in the Solar System for astrobiology, but the 44-minute one-way light delay to Jupiter, Jupiter's intense radiation environment, and a 90 K cryogenic surface make it one of the most demanding environments to operate a robot in. Every decision from landing onward must be made onboard; there is no real-time human-in-the-loop safety net.

The rover is a four-wheel, differential-drive vehicle riding on a single-pivot rocker, sized against a 147.25 kg contingency mass cap, and powered by a Radioisotope Thermoelectric Generator (RTG) rather than solar arrays, since solar irradiance at Europa's distance from the Sun collapses to roughly 50 W/m$^2$. Its primary payload is a REASON-heritage dual-frequency, ice-penetrating radar sounder, the same instrument family currently flying aboard NASA's Europa Clipper spacecraft [7]. Where Clipper collects wide-swath, low-resolution orbital snapshots of the ice shell, this mission's rover drives a continuous surface traverse, building a spatially registered subsurface profile at 100 m resolution or better and autonomously flagging the thinnest viable ice for a future cryobot deployment.

Two orbital agents support the rover from a 200 km circular Europa orbit, providing terrain mapping ahead of the traverse, a UHF proximity relay for the rover's science and command traffic, and an independent ranging source the rover can use to recover its position after a localization fault. Communication with Earth exists only as a low-rate contingency beacon; a direct link at Europa's distance closes at roughly 7 bits per second, which is why the relay architecture is not optional but foundational to the mission design.

Across a 30-sol surface mission, the system is expected to produce a continuous, georeferenced subsurface ice-thickness profile along the traverse track, a validated candidate cryobot insertion site (ice thickness under 5 km, surface slope under 20°, terrain roughness under 0.7), and a full science and engineering telemetry archive suitable for both real-time onboard decision-making and post-mission analysis by Earth-based planetary scientists. The requirements, environmental analysis, concept of operations, payload rationale, data strategy, and subsystem designs that support this objective are detailed in the sections that follow.

# 2 Research Context and Gap Framing

Three converging research gaps motivate this mission, drawn from the OWL Planetary Decadal Survey, a NASA STMD technology shortfall, and a NASA SBIR/STTR solicitation topic.

## 2.1 OWL Decadal Survey: Ocean Worlds Exploration & Sampling

**Topic:** Ocean Worlds Exploration & Sampling [1, pp. 36–39]

**Interest:** Ocean worlds like Europa represent the most compelling environments for studying life beyond Earth, and the most difficult for autonomous systems. What draws me to this topic is the engineering challenge that arises from the question: how do you keep a robot operational, navigating, and decision-making in a GPS-denied, high-radiation, cryogenic environment with no possibility of real-time human intervention? The decadal survey speaks to the real physical needs here: Jupiter's Galilean moons reveal "the fundamental importance of gravitational tides in generating internal heat," with those tides actively "melting ice on Europa and Ganymede" [1, p. 37]. That subsurface ocean exists because of a process we cannot turn off, and any robotic system sent to study it must be designed around that reality.

**Connection to Space Exploration:** Accessing a subsurface ocean requires penetrating kilometers of ice, a process that may yield in-situ resources (liquid water, chemical energy gradients) critical to sustaining any long-duration mission. Understanding the composition of that ocean directly informs the feasibility of using local materials to support future human or robotic infrastructure in the outer solar system.

**Relationship to Robotics:** Reaching and operating in a subsurface ocean demands a full robotic stack operating without outside reference: specialized robots for ice penetration, underwater vehicles for ocean navigation, and autonomous sampling systems capable of detecting and responding to scientifically interesting features. Every subsystem (perception, localization, path planning, controls) must function independently.

## 2.2 NASA STMD Shortfall: Multi-Agent Robotic Coordination

**Topic:** Multi-Agent Robotic Coordination and Interoperability for Cooperative Task Planning and Performance [2]

**Interest:** Swarm robotics is the next step in advancing the capabilities of robots to complete high-level missions. NASA's own shortfall rankings confirm how urgent this is: multi-agent coordination received an integrated score of 4.7856 and ranked 139th among all civil space technology gaps [2], placing it right in the near-term priority tier for Autonomous Systems and Robotics. I am most interested in the version of it where each robot in the swarm is genuinely capable on its own, not a simple unit that only functions within the group, but a system of robots with unique capabilities that all come together to complete a high-level task.

**Connection to Space Exploration:** Cooperative exploration and mapping by a distributed robotic fleet dramatically increases the spatial coverage and fault tolerance of resource missions. A swarm can survey a large asteroid surface, map subsurface ice deposits via distributed sensing, or maintain constant monitoring of a volatile-rich region far more efficiently than any single agent, with adversarial identification algorithms when individual units fail.

**Relationship to Robotics:** This topic is a direct application of the consensus and coordination algorithms I have studied for the last year. Deploying them in space introduces new constraints (limited bandwidth, high latency, radiation-induced faults, and no GPS) that stress-test the theoretical properties I've studied (convergence, resilience, scalability).

## 2.3 NASA SBIR/STTR Topic: Flight Dynamics and Navigation Technologies

**Topic:** Flight Dynamics and Navigation Technologies [3] (Topic COMNAV.1.S26B)

**Interest:** Communication delays and blackouts are not random edge cases in deep-space missions; they are the default operating condition. The SBIR solicitation states the gap directly: "NASA currently has limited navigational and guidance technologies that permit fully autonomous approach, proximity operations, distributed spacecraft operations, and landing without support from Earth-based resources" [3]. This shortfall interests me because it forces onboard autonomy to be the primary decision-making layer, eliminating the safety net of human-in-the-loop oversight.

**Connection to Space Exploration:** Remote resource operations (prospecting, extraction, processing) cannot depend on continuous communication to Earth. Whether managing an ISRU plant on the lunar surface or conducting a sampling traverse on an asteroid, the system must detect anomalies, replan, and execute without waiting for human feedback.

**Relationship to Robotics:** My research in consensus algorithms and resilient multi-agent coordination already deals with how distributed systems reach agreement under adversarial conditions. Extending this to communication-constrained space contexts is a natural progression: the same algorithms that allow a swarm of Crazyflies to maintain stability during packet loss should also apply to a team of planetary rovers operating through a communication outage.

# 3 Mission Definition and Environment

## 3.1 Mission Objective & Destination

**Selected Research Gap:** While most robotic space missions have focused on a single agent carrying all the capabilities needed to complete the mission objectives, the future of advanced planetary exploration lies in multi-agent systems that are able to share responsibility and support each other. In order to fully understand a body as complex as Europa, it is necessary to divide surveying tasks across the main exploration rover and the orbital agents needed to scout terrain ahead of it. This mission directly addresses three converging research gaps: autonomous operations in GPS-denied, high-radiation environments [1]; multi-agent coordination and interoperability for cooperative task planning [2]; and fully autonomous approach and proximity operations without Earth-based navigational support [3].

**Mission Statement:** The multi-agent Europa surface system shall autonomously characterize the ice shell and identify a cryobot insertion site, operating without real-time Earth support.

**Success Criteria (summary):** Full traverse track covered at $\leq 100$ m along-track spacing, ice-thickness accuracy within $\pm 10\%$ against calibration targets, and at least one candidate insertion site flagged by the onboard algorithm (ice thickness $< 5$ km, surface slope $< 20°$, terrain roughness $< 0.7$). Section 3.3 develops the full mission- and operations-level requirements that this criterion is built from.

**Destination Justification:** My mission takes place on Europa, Jupiter's icy moon. I chose this destination due to the abundance of knowledge there is to gain from an in-depth exploration of the system. Europa holds strong indicators of conditions favorable to life and contains all the chemical building blocks necessary to support it [1]. The layers of ice introduce in-situ resources that might benefit further exploration of the moon as long-term infrastructure is developed, as well as imposing significant rover design constraints. Any surface vehicle must feature traction-optimized wheels capable of accurately traversing the fractured, chaotic terrain.

Europa is also incredibly challenging. The 44-minute one-way light delay means every decision from landing onward must be fully autonomous [3]. Due to the intense radiation environment driven by Jupiter's magnetosphere, the rover relies on orbiting satellite robots to scan the terrain and share terrain maps prior to surface traversal. Such a system has to be highly robust to faulted or adversarial agents, as incorrect or incomplete maps shared across the swarm are likely to produce dangerous navigation decisions. This location is essential to the mission because it provides an incredibly challenging environment that necessitates smart, advanced robot design and algorithm development, the perfect stage to push robotic capabilities to their limits.

### 3.2 Environmental & Operational Challenges

Europa's environment is hostile in ways that shape every design decision in this mission. Jupiter's magnetosphere floods the moon's surface with high-energy charged particles, exposing electronics to radiation doses far greater than what commercial components can tolerate. This motivates a requirement for radiation-hardened electronics and a shielded vault setup for the rover's core computing hardware, similar to the approach taken on the Galileo spacecraft [1]. Redundancy at the agent level is an equally important mitigation: if one surface relay node builds up enough single-event upsets to go silent, the remaining agents must autonomously reconfigure the network to maintain coverage.

Surface temperatures on Europa are around 90 K, which introduces thermal management challenges for both batteries and actuators. Power storage capacity drops significantly at cryogenic temperatures, meaning the rover must actively manage its thermal systems to keep cells within an operational range rather than simply relying on passive insulation. Solar power is not viable at Europa's distance from the Sun (irradiance collapses to roughly 50 W/m$^2$), so the system instead relies on an RTG.

The complete absence of GPS and the communication blackout imposed by the 44-minute light delay mean the swarm cannot depend on any external reference for localization or replanning. Each agent must maintain its own state estimate using onboard sensors and inter-agent ranging, and the swarm must reach consensus on shared map data using algorithms that are resilient to packet loss and radiation-induced faults [2, 3]. There is no human in the loop to catch a bad map before the rover commits to a traverse. Table 1 summarizes how each environmental driver maps onto a subsystem-level design response; the full derivation appears in the subsystem sections of Section 7.

| **Environmental Driver** | **Operational Consequence** | **Design Response** |
|---|---|---|
| Jupiter magnetosphere radiation ($\geq$ 100 krad) | Electronics degradation, single-event upsets | Radiation vault (Ti-6Al-4V + Pb/poly liner), agent-level redundancy |
| 44-min one-way light delay | No real-time human command | Full onboard autonomy, consensus-based multi-agent decisions |
| No GPS, no magnetic field | No absolute position/heading reference | Visual odometry + IMU + fine sun sensors + inter-agent ranging |
| 90 K cryogenic surface | Battery capacity loss, actuator lubricant stiffening, brittle materials | MLI + RTG waste-heat routing + survival heaters, pre-motion warm-up |
| $\sim$ 50 W/m$^2$ solar irradiance | Solar arrays not viable | RTG primary power source |
| Fractured, chaotic ice terrain | Traction/mobility risk, hazard density | Cleated wheels, onboard hazard rejection, low-speed arc-based traversal |

Table 1: Environmental drivers and their design responses.

### 3.3 Mission Success Criteria

Requirements are drawn from the master requirements spreadsheet (Appendix B contains the full set, REQ.01.01 through REQ.12.04). Table 2 presents three mission-level requirements and Table 3 presents three operations-level requirements, each with its justification, verification approach, and associated risk.

| ID | Statement | Justification | Verification / Risk |
|---|---|---|---|
| REQ.01.01 | The multi-agent Europa system shall operate fully autonomously for the entire surface mission duration without requiring real-time command uplinks from Earth. | The 44-minute one-way light delay makes real-time commanding physically impossible; all decisions must be onboard. | 72-hr hardware-in-the-loop mission sequence with a simulated blackout, zero unresolved Earth dependencies. Risk-01: single-point autonomy failure with no fallback human override. |
| REQ.01.02 | The system shall survive and maintain operational status after exposure to a total ionizing dose of no less than 100 krad over the nominal mission lifetime. | Jupiter's magnetosphere delivers doses far exceeding near-Earth missions; Europa Clipper's radiation vault targets $\sim 100$ krad as its design threshold. | Co-60 gamma irradiation of radiation-sensitive components to 110 krad (10% margin) with functional monitoring throughout. RISK-02: accumulated SEUs degrade consensus convergence before mission completion. |
| REQ.01.03 | The system shall maintain a minimum inter-agent network connectivity of two active relay nodes at all times during surface operations to ensure consensus is achievable. | Consensus algorithms require a minimum connected graph; two relays plus the rover guarantee a 3-node connected topology, the minimum viable configuration. | Network fault simulation with progressive relay failures; reconfiguration must trigger before dropping below 2 active nodes. RISK-03: simultaneous radiation-induced failure of multiple relay nodes isolating the rover. |

Table 2: Mission-level requirements.

| ID | Statement | Justification | Verification / Risk |
|---|---|---|---|
| REQ.02.01 | The surface robot shall maintain a localization accuracy of no worse than ±0.5 m relative to its last known position during traverse segments up to 50 m in the absence of GPS. | Cryobot insertion site selection requires meter-level positioning to match the radar-identified thin-ice candidate; consistent with visual odometry performance on Mars rovers. | Analog-terrain field test, 50 m commanded segments, onboard estimate vs. ground truth across 20 runs. RISK-04: ice surface texture homogeneity reducing visual odometry feature tracking. |
| REQ.03.01 | The ground-penetrating radar sounder shall resolve ice shell thickness to within ±10% accuracy for ice layers between 1 km and 30 km depth, producing a subsurface map at a minimum spatial resolution of 100 m along the traverse track. | Identifying the optimal cryobot insertion site requires knowing where the ice shell is thinnest; consistent with Europa Clipper REASON design parameters. | Laboratory radar range measurements against known-thickness ice-analog targets across the full depth range. RISK-05: ice-shell heterogeneity or brine inclusions creating radar clutter. |
| REQ.04.01 | The inter-agent mesh network shall achieve consensus on a shared terrain map update within no more than 10 minutes under a degraded network condition of up to 30% packet loss on any single inter-agent link. | Traverse planning depends on a current, consensus-validated terrain map; a 10-minute bound keeps planning delay acceptable, and 30% loss is a plausible radiation-induced degradation scenario. | Software-in-the-loop consensus simulation, 30% packet loss injected on every link across 100 test runs. RISK-06: radiation-induced burst errors exceeding 30% loss on multiple links simultaneously, stalling consensus. |

Table 3: Operations-level requirements.

# 4 Concept of Operations (ConOps)

## 4.1 Overview Narrative

The multi-agent system deployed to Europa's surface autonomously characterizes ice shell thickness and identifies an optimal cryobot insertion site without real-time Earth support. The destination is Europa, chosen because its subsurface ocean makes it one of the most promising locations in the Solar System for astrobiology and future exploration. The mission deploys a surface rover equipped with a ground-penetrating radar sounder, supported by two orbital reconnaissance agents that provide terrain mapping, communications relay, and cooperative localization. By leveraging radar sounding techniques analogous to the Europa Clipper REASON instrument, the system generates detailed subsurface ice maps to support future cryobot missions.

The mission consists of six main phases: Launch and Commissioning, Cruise and Approach, Europa Orbit Insertion and Reconnaissance, Landing/Checkout and Site Characterization, Surface Survey and Ice Characterization, and Data Transmission and End-of-Mission Operations (matching the six phase rows in Table 4). Launch and cruise prepare the system for arrival at Europa. Upon arrival, the orbital agents conduct reconnaissance and identify a suitable landing region. Landing is followed by rover deployment, system checkout, and sensor calibration. Main surface operations focus on autonomous rover traversal while collecting radar measurements of the ice shell and building a subsurface map; the orbital agents continuously update terrain maps and relay communications. The final phase includes science data downlink, candidate cryobot site selection, optional extended operations, and safe system passivation.

The rover operates autonomously during critical events such as landing, navigation, radar sounding, and site-selection activities. The two orbital agents provide terrain reconnaissance, communications relay, cooperative localization support, and validation of shared mapping data. Ground teams remain in the loop during cruise, strategic mission planning, and anomaly resolution, but all tactical decisions must be performed onboard due to the approximately 43–44 minute one-way communication delay between Earth and Jupiter. Radiation exposure, cryogenic temperatures, communication delays, and limited bandwidth are the key operational constraints that shape every phase below.

## 4.2 Operational Timeline

Table 4 presents the mission-phase-level ConOps timeline (mission phases, operation modes, and key events), and Figure 1 presents the corresponding operational flow graphic. A finer-grained, minute-by-minute breakdown for the first three sols of surface operations plus the steady-state daily cycle and end-of-mission sequence is maintained in the source spreadsheet (*2.4 ConOps Detailed Timeline*); it is summarized in Table 5 rather than reproduced row-by-row here for length.

| **Time** | **Mode** | **Objective / Milestone** | **Op. Mode** | **Risk** | **Duration** |
|---|---|---|---|---|---|
| Launch day, 0–1 | Launch & Commissioning | Spacecraft separation, initial health checks | Scripted/Auto | High | 1 day |
| Cruise, yrs 0–6 | Cruise & Approach | Maintain health, trajectory correction maneuvers | Semi-Auto | Medium | ∼6 years |
| Europa arrival | Orbit Insertion & Reconnaissance | Deploy orbital agents, identify landing region | Autonomous | High | 30 days |
| Surface day 0–5 | Landing, Checkout & Site Characterization | Deploy rover, verify health, calibrate payload | Autonomous | High | 5 days |
| Surface day 5–28 | Surface Survey & Ice Characterization | Map subsurface ice, identify candidate cryobot sites | Autonomous | Medium | 23 days |
| Surface day 28–30 | Data Transmission & End of Mission | Downlink archive, select final site, passivate | Semi-Auto | Low | 2 days |

Table 4: Mission-phase-level Concept of Operations timeline.

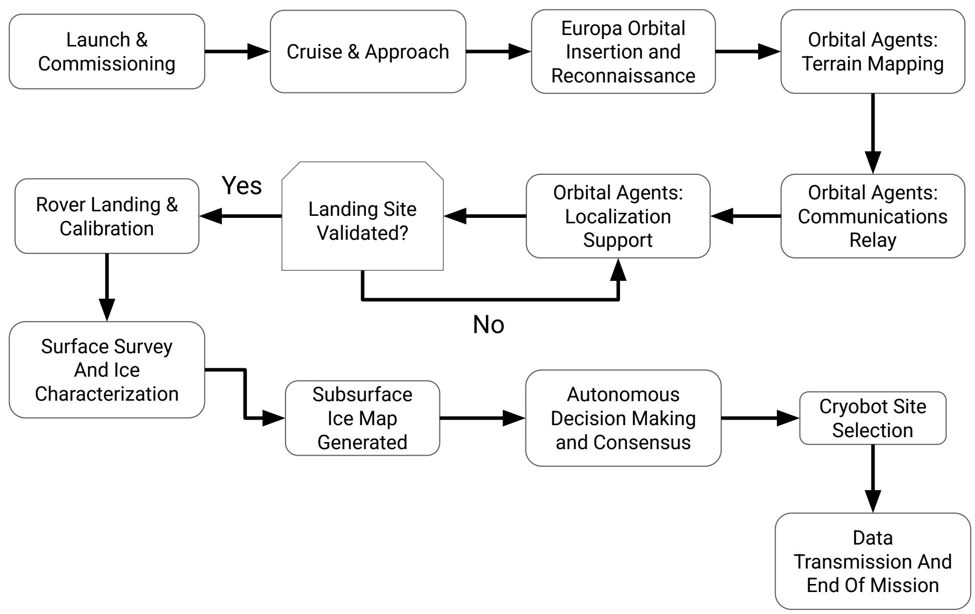


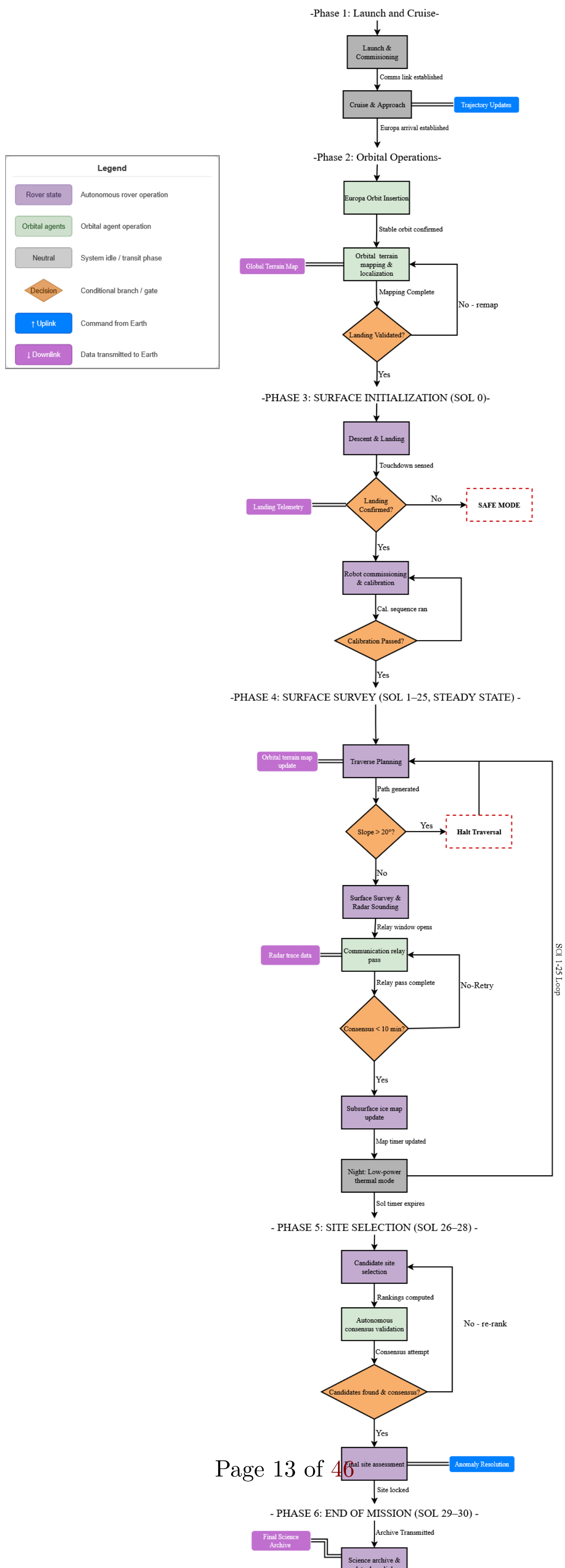

Figure 1: Operational flow across the six mission phases, including the landing-site validation decision point and the autonomous consensus loop that drives cryobot site selection.

| **Sol(s)** | **Phase** | **Key Events** |
|---|---|---|
| 0 | Initialization | Descent/landing (0–30 min); commissioning and health checks; orbital comms verification; localization initialization; radar payload init, calibration, and validation; initial site characterization; night low-power thermal management. |
| 1–2 | Activation | Daily cycle established: autonomous traverse planning, orbital terrain mapping update, surface survey blocks, localization verification, radar data collection, communications relay pass, subsurface ice map generation. |
| 3–25 | Steady-state operations | Repeating daily cycle: traverse planning → survey operations → radar data collection → communications relay pass → subsurface ice map update → night low-power thermal management. |
| 26–28 | Site selection | Candidate site ranking and autonomous consensus validation (repeated over two sols), final site assessment, cryobot site selection. |
| 29 | Archive & validation | Final science archive generation, final terrain map validation. |
| 30 | End of mission | Final data dump, rover/payload safing, communications shutdown, passivation and final health beacon. |

Table 5: Summary of the minute-by-minute surface operations timeline (source: 2.4 ConOps Detailed Timeline spreadsheet).

### 4.3 Flight Software State Diagram

Figure 2 presents the flight-software state flow (source: *2.5 ConOps State Flow Diagram*), covering all six mission phases from Launch and Cruise through End of Mission. Each rover-state block (purple) is bounded by an orbital-agent or ground-command dependency where one exists (e.g., Orbital Terrain Mapping & Localization gates Landing Validated? before Phase 3 begins), and every conditional branch resolves to a named exit rather than an implicit fall-through: a failed landing confirmation routes to Safe Mode, an out-of-tolerance slope check during Phase 4 traverse planning routes to Halt Traversal, and a sub-10-minute consensus failure on the relay pass retries rather than proceeding on stale data (REQ.04.01). The Phase 4 steady-state block (Traverse Planning → Survey/Sounding → Slope Check → Relay Pass → Consensus Check → Map Update → Night Low-Power) loops for sols 1–25 before handing off to Phase 5 site selection, matching the daily cycle in Table 5.


Legend
Rover state
Autonomous rover operation
Orbital agents
Orbital agent operation
Neutral
System idle / transit phase
Decision
Conditional branch / gate
↑ Uplink
Command from Earth
↓ Downlink
Data transmitted to Earth
-Phase 1: Launch and Cruise-
Launch & Commissioning
Comms link established
Cruise & Approach
Trajectory Updates
Europa arrival established
-Phase 2: Orbital Operations-
Europa Orbit Insertion
Stable orbit confirmed
Orbital terrain mapping & localization
Global Terrain Map
Mapping Complete
No - remap
Landing Validated?
Yes
-PHASE 3: SURFACE INITIALIZATION (SOL 0)-
Descent & Landing
Touchdown sensed
Landing Confirmed?
Landing Telemetry
No
SAFE MODE
Yes
Robot commissioning & calibration
Cal. sequence ran
Calibration Passed?
Yes
-PHASE 4: SURFACE SURVEY (SOL 1–25, STEADY STATE) -
Orbital terrain map update
Traverse Planning
Path generated
Slope > 20°?
Yes
Halt Traversal
No
Surface Survey & Radar Sounding
Relay window opens
Radar trace data
Communication relay pass
Relay pass complete
No-Retry
Consensus < 10 min?
Yes
Subsurface ice map update
Map timer updated
Night: Low-power thermal mode
SOL 1-25 Loop
Sol timer expires
- PHASE 5: SITE SELECTION (SOL 26–28) -
Candidate site selection
Rankings computed
Autonomous consensus validation
No - re-rank
Consensus attempt
Candidates found & consensus?
Yes
Final site assessment
Anomaly Resolution
Site locked
- PHASE 6: END OF MISSION (SOL 29–30) -
Archive Transmitted
Final Science Archive
Science archive &

# 5 Payload-Driven Design

## 5.1 Key Payload Description

**Payload: REASON (Radar for Europa Assessment and Sounding: Ocean to Near-surface)**

REASON is a dual-frequency ice-penetrating radar sounder currently flying aboard NASA's Europa Clipper spacecraft. The instrument was developed by researchers at the University of Texas Institute for Geophysics and fabricated by JPL, and it launched on October 14, 2024 atop a SpaceX Falcon Heavy rocket bound for Europa [6]. REASON operates at two frequency bands simultaneously, allowing it to distinguish between near-surface structure and deep basal reflections, a capability no previous Europa-targeted instrument has demonstrated [4]. The instrument is flight-proven hardware with full mission-level heritage, placing it at TRL 9.

Europa Clipper carries REASON as part of a flyby science campaign: the spacecraft loops past Europa roughly 50 times, collecting wide-swath regional maps of the ice shell from altitude [5]. This mission seeks to address what Clipper does not cover: surface-level, high-resolution traverse profiling targeted at a specific candidate cryobot insertion site. There is no rover, no in-situ localization, and no onboard site-selection logic on Clipper, and the orbital geometry limits any single flyby to a brief snapshot rather than a continuous along-track profile. This mission is essentially a continuation of Clipper: a surface rover carrying an adapted REASON-heritage sounder drives a continuous traverse, building a spatially registered subsurface map at 100 m resolution or better, and autonomously flags the thinnest viable ice for a future cryobot deployment.

REASON measures four primary parameters. Two-Way Travel Time (ns) is the elapsed time between the transmitted pulse and the received basal reflection, and serves as the raw measurement. Signal Amplitude (dB) captures the strength of the return from the ice-ocean interface; echoes that fall into the noise floor are flagged to prevent false positives in site selection. Bulk Ice Dielectric Permittivity (dimensionless) is estimated from the sounding's travel characteristics ($\varepsilon_r \approx 3.15$ for pure water ice), with deviations flagging brine pockets or structural impurities. Finally, Derived Ice Shell Thickness (m) is computed autonomously onboard using $d = c \cdot t \,/\, (2\sqrt{\varepsilon_r})$, where $c$ is the speed of light.

There are two crucial users of REASON's data. The onboard autonomous planning algorithm is the real-time consumer: it continuously processes daily trace summaries to locate viable cryobot deployment sites (ice thickness $< 5$ km, surface slope $< 20°$, roughness index $< 0.7$) and issues a deployment trigger without human intervention when conditions are satisfied. Earth-bound planetary scientists form the long-term consumer base: the full downlinked dataset provides a high-resolution, spatially registered along-track profile of Europa's ice shell that can be used to model global habitability, constrain ocean-access scenarios, and plan future deep-subsurface entry missions.

The rover also carries a secondary MEDA-class environmental monitoring suite, adapted from the heritage instrument flying aboard NASA's Perseverance rover [16], to log local surface-environment context (temperature, pressure, dust/radiation proxies) alongside each radar trace (Table 12). It supports interpretation of the REASON-heritage sounder's returns but does not itself drive a Section 5 requirement.

## 5.2 Payload Requirements

The payload imposes concrete, traceable requirements on every other subsystem. Pointing: REQ.03.02 requires the antenna boresight to stay within $\pm 5°$ of nadir during active sounding, forcing the rover to halt traversal whenever surface slope exceeds $20°$; this requirement is carried by

Structures (antenna mast/mounting geometry) and Mobility (slope-triggered halt logic). Power: REQ.03.03 requires a dedicated peak power allocation for radar transmission pulses and commands drive-motor shutdown during each sounding window, since the RTG's fixed continuous output cannot support simultaneous driving and sounding without risking under-powering the radar pulse. Data handling: REQ.03.04 requires the CDH subsystem to retain a minimum of 7 sols of uncompressed radar trace data and apply lossless compression at a minimum 2:1 ratio before inter-agent relay transmission, since raw science data accumulates faster than the non-deterministic Earth relay link can clear it. Thermal: REQ.05.01 requires the radar electronics (and all avionics) to stay within 0–40°C during active operations and above −20°C in night survival mode. Together these four requirements are why the payload, rather than a generic bus design, drove the rover's mobility cadence (Section 7.7), power budgeting (Section 7.4), onboard storage strategy (Section 6), and thermal architecture (Section 7.5).

# 6 Data Strategy and Deliverables

## 6.1 Data Product Table

**Mission Objective:** Autonomously characterize Europa's ice shell thickness along a surface traverse and identify an optimal cryobot insertion site without real-time Earth support (REQ.03.01: ground-penetrating radar sounder resolves ice thickness to ±10% for depths of 1–30 km at $\geq 100$ m along-track spatial resolution).

The data product is organized into four sheets so the dataset stays clear and usable. Table 6 summarizes each; full field-by-field definitions live in the source 2.1 Data Product deliverable.

| Sheet | Key Fields (units) | Intended Consumers |
|---|---|---|
| Radar_Traces | Trace ID, sol, along-track position (m), two-way travel time (ns), amplitude (dB), permittivity (–), derived ice thickness (m), quality flag | Onboard site-selection algorithm (real time); planetary scientists (post-mission) |
| Localization_Log | Position X/Y (m, local ENU), heading (deg), $1\sigma$ uncertainty (m), REQ.02.01 error flag | GNC/onboard planner; mission operations engineers |
| Terrain_and_Health | Surface slope (deg), roughness index (0–1), surface temperature (K), accumulated TID (krad), radar noise floor (dB), inter-agent packet loss (%), per-trace quality flag | Onboard hazard/site filter; systems engineers monitoring health trends |
| Daily Summary | Traverse distance/sol (m), valid trace count, minimum ice thickness/sol (m), candidate site count, mean consensus convergence time (min) | Mission planners (per-sol go/no-go); operations team |

Table 6: Data product sheets, key fields, and intended consumers.

Each data category maps to a specific requirement. The radar sounder returns are the direct measurement required by REQ.03.01; without travel time and permittivity there is no thickness inversion and no subsurface map. The localization log is required by REQ.02.01: each trace must be placed within ±0.5 m to produce a spatially accurate map rather than an unordered list of depth readings. Terrain context drives the site-selection filter, where slope and roughness constrain physical accessibility independent of ice thickness. System health data is required by REQ.01.02 and REQ.04.01: elevated TID can degrade radar performance, and packet loss above 30% directly stresses the consensus convergence bound. The main known limitation is permittivity estimation uncertainty when brine inclusions or heterogeneous layering are present (RISK-05); brine lenses increase local permittivity and can cause the thickness algorithm to underestimate depth to the basal interface. The `CLUTTER` quality flag mitigates this by excluding affected traces from site selection, but does not resolve the ambiguity without a secondary instrument.

## 6.2 Example Output

On Sol 1 at 94.4 s elapsed mission time (trace `TR-0001`), REASON recorded a two-way travel time of 21,647.3 ns, a return amplitude of −41.62 dB, a bulk ice permittivity of 3.06, and a derived ice shell thickness of 1,855.5 m at rover position (93.8 m east, −7.1 m north), with an accumulated total ionizing dose of 0.542 krad and the record flagged as `NOMINAL`. Tables 7 and 8 present this trace together with the following 19 consecutive Sol-1 traces, exported directly from the *Data Product Space Robotics* spreadsheet (Radar_Traces sheet, rows `TR-0001` through `TR-0020`); four of the twenty are flagged `NOISE_FLOOR` or `CLUTTER` and correctly excluded from site-selection logic per Section 6.1.

| Trace ID | Sol | Pos. E (m) | Pos. N (m) | TWT (ns) | Ampl. (dB) | $\varepsilon_r$ |
|---|---|---|---|---|---|---|
| TR-0001 | 1 | 93.8 | −7.1 | 21,647.3 | −41.62 | 3.06 |
| TR-0002 | 1 | 185.5 | −3.2 | 169,839.5 | −45.53 | 3.10 |
| TR-0003 | 1 | 296.4 | 11.4 | 164,535.9 | −25.34 | 3.19 |
| TR-0004 | 1 | 382.9 | 7.0 | 131,950.8 | −31.93 | 3.34 |
| TR-0005 | 1 | 484.3 | −0.6 | 95,357.8 | −49.20 | 3.46 |
| TR-0006 | 1 | 577.3 | 7.4 | 23,845.1 | −27.13 | 3.40 |
| TR-0007 | 1 | 693.8 | 9.4 | 140,352.9 | −51.49 | 3.25 |
| TR-0008 | 1 | 779.5 | −1.4 | 145,118.1 | −38.44 | 3.37 |
| TR-0009 | 1 | 874.0 | 13.5 | 38,346.4 | −40.26 | 3.38 |
| TR-0010 | 1 | 960.4 | 18.9 | 118,968.0 | −44.61 | 3.30 |
| TR-0011 | 1 | 1,062.5 | 32.2 | 133,729.9 | −55.42 | 3.44 |
| TR-0012 | 1 | 1,156.1 | 26.0 | 41,835.8 | −59.87 | 3.36 |
| TR-0013 | 1 | 1,266.6 | 26.3 | 32,728.3 | −34.99 | 3.42 |
| TR-0014 | 1 | 1,361.5 | 16.1 | 181,816.0 | −23.10 | 3.46 |
| TR-0015 | 1 | 1,446.0 | 12.0 | 187,416.6 | −27.73 | 3.12 |
| TR-0016 | 1 | 1,529.2 | 18.9 | 157,620.1 | −20.88 | 3.27 |
| TR-0017 | 1 | 1,647.0 | 8.9 | 106,870.6 | −35.74 | 3.48 |
| TR-0018 | 1 | 1,754.6 | 15.3 | 84,222.6 | −33.13 | 3.19 |
| TR-0019 | 1 | 1,844.6 | 2.2 | 17,702.0 | −37.84 | 3.29 |
| TR-0020 | 1 | 1,926.9 | −10.8 | 19,528.7 | −46.78 | 3.26 |

Table 7: Radar_Traces sample export, Sol 1, traces TR-0001–TR-0020: position and raw sounding returns.

| Trace ID | Thickness (m) | TID (krad) | Flag |
|---|---|---|---|
| TR-0001 | 1,855.5 | 0.542 | NOMINAL |
| TR-0002 | 14,478.5 | 0.577 | NOMINAL |
| TR-0003 | 13,818.1 | 0.735 | NOMINAL |
| TR-0004 | 10,827.1 | 0.817 | NOISE_FLOOR |
| TR-0005 | 7,686.7 | 0.999 | NOMINAL |
| TR-0006 | 1,939.1 | 1.09 | NOMINAL |
| TR-0007 | 11,678.8 | 1.258 | NOMINAL |
| TR-0008 | 11,851.5 | 1.333 | NOMINAL |
| TR-0009 | 3,129.6 | 1.496 | NOMINAL |
| TR-0010 | 9,826.6 | 1.577 | CLUTTER |
| TR-0011 | 10,811.6 | 1.725 | NOMINAL |
| TR-0012 | 3,423.0 | 1.842 | NOMINAL |
| TR-0013 | 2,654.3 | 1.941 | NOMINAL |
| TR-0014 | 14,663.4 | 2.07 | NOMINAL |
| TR-0015 | 15,916.3 | 2.154 | NOMINAL |
| TR-0016 | 13,081.8 | 2.263 | CLUTTER |
| TR-0017 | 8,590.6 | 2.463 | CLUTTER |
| TR-0018 | 7,077.0 | 2.58 | NOMINAL |
| TR-0019 | 1,463.0 | 2.611 | NOMINAL |
| TR-0020 | 1,623.1 | 2.758 | NOMINAL |

Table 8: Radar_Traces sample export, Sol 1, traces TR-0001–TR-0020: derived ice thickness, accumulated dose, and quality flag.

Table 9 exports both rows currently populated in the Daily_Summary sheet (Sol 1–2); the source spreadsheet has not yet been extended past Sol 2, so this is the complete Daily_Summary export available rather than an arbitrary 2-row excerpt.

| Sol | Traverse (m) | Valid | Bad | Min. Thickness (m) | Cand. Sites | Pkt. Loss (%) | Consensus (min) |
|---|---|---|---|---|---|---|---|
| 1 | 1,841.6 | 16 | 4 | 1,463.0 | 7 | 19.845 | 5.969 |
| 2 | 1,804.8 | 16 | 4 | 9,750.0 | 0 | 19.795 | 5.959 |

Table 9: Daily_Summary sample export, Sol 1–2 (complete export of the sheet's current contents).

The Sol-1 minimum-thickness and candidate-site figures above are recomputed directly from the Radar_Traces sample in Tables 7–8: the minimum is 1,463.0 m (`TR-0019`), the lowest thickness among the sol's 16 `NOMINAL`-flagged traces, and the candidate count of 7 is the subset of those traces under the 5 km thickness screen (Section 6.1); this is an upper bound on true candidates pending the slope/roughness fields, which live in the Terrain_and_Health sheet and are not part of this trace export. Sol 2's minimum thickness of 9,750.0 m sits above that same 5 km screen, so the correct candidate count for Sol 2 is 0: the screen is correctly rejecting a sol whose ice is uniformly too thick, not producing a spurious candidate. Sol-2 trace-level records are not yet present in the source spreadsheet, so the 9,750.0 m minimum itself could not be cross-checked against raw traces the way Sol 1's was, but the reported candidate count is consistent with it either way.

# 7 System and Subsystem Designs

## 7.1 Functional & Nonfunctional Requirements

Table 10 and Table 11 highlight five functional and five nonfunctional requirements drawn from the master requirements set (Appendix B has the complete REQ.01.01–REQ.12.04 list with sources, subsystem ownership, and V&V plans).

| ID | Functional Requirement (what the system must do) |
|---|---|
| REQ.02.01 | Maintain localization accuracy no worse than ±0.5 m over any 50 m traverse segment in the absence of GPS. |
| REQ.03.01 | Resolve ice shell thickness to within ±10% for 1–30 km depths, at $\geq$ 100 m along-track resolution. |
| REQ.06.01 | Autonomously plan and execute $\geq$ 100 m of surface traverse per sol, pausing at $\leq$ 100 m intervals for a radar sounding stop, and resume without ground uplink. |
| REQ.11.02 | Assess terrain out to $\geq$ 8 m ahead of each drive arc and autonomously reject any arc with a step $>$ 0.20 m, slope $>$ 20°, or roughness $>$ 0.7. |
| REQ.11.04 | Halt traverse and enter a navigation hold whenever position uncertainty exceeds 0.5 m ($1\sigma$), and re-initialize via relay ranging plus map correlation. |

Table 10: Representative functional requirements.

| ID | Nonfunctional Requirement (quality attribute / constraint) |
|---|---|
| REQ.01.01 | Operate fully autonomously for the entire surface mission without real-time Earth uplinks (dependability / autonomy constraint). |
| REQ.01.02 | Survive and remain operational after $\geq$ 100 krad total ionizing dose over the mission lifetime (survivability). |
| REQ.05.01 | Maintain avionics/battery temperature within 0–40°C active, $>$ $-20$°C night mode (environmental performance). |
| REQ.08.01 | Generate $\geq$ 110 We (BOL) / $\geq$ 80 We continuous via MMRTG-class RTG, $\geq$ 1.9 kWh usable per sol (capacity / performance). |
| REQ.10.03 | Close the rover–relay proximity link with $\geq$ 3 dB margin over the required $E_b/N_0$ at the 588 km worst-case range (reliability / performance margin). |

Table 11: Representative nonfunctional requirements.

## 7.2 Robot System Overview

The system is a three-agent architecture: one surface rover and two orbital relay agents in 200 km circular Europa orbits. Broadly, the rover carries the science payload (REASON-heritage GPR sounder) and mobility/GNC stack needed to physically reach and characterize a candidate cryobot site; the orbital agents carry the terrain-mapping and communications-relay capability the rover cannot provide for itself in a GPS-denied environment. Together they fulfill the mission objective end to end: the orbiters find where it is safe to look, the rover determines exactly how thick the ice is there, and the mesh network gets that answer back to Earth despite a 44-minute light delay.

Figure 3 shows how the rover's subsystems interface with one another. Power (RTG + battery packs) feeds every other subsystem over regulated 28 V and 5 V buses through the PDU/EPS board; Thermal routes RTG waste heat to the radiators and survival heaters and isolates the radiator loop with MLI and thermal switches; Structures provides the physical mounting, radiation vault, and load path that the deployed legs, mast, and antenna assembly all attach to; Avionics (the RAD750-class flight computer and I/O boards) hosts the GNC compute stack and mediates between Comms, Payload, and Mobility; and Comms carries science data out over UHF to the relay agents and commands back in, with an X-band link reserved as a low-rate, direct-to-Earth contingency beacon. The GNC compute stack (visual odometry → navigation EKF → hazard assessment → arc path planner → arc controller → mode manager/FDIR) is the software layer that actually closes the loop between all of these physical interfaces every drive arc.

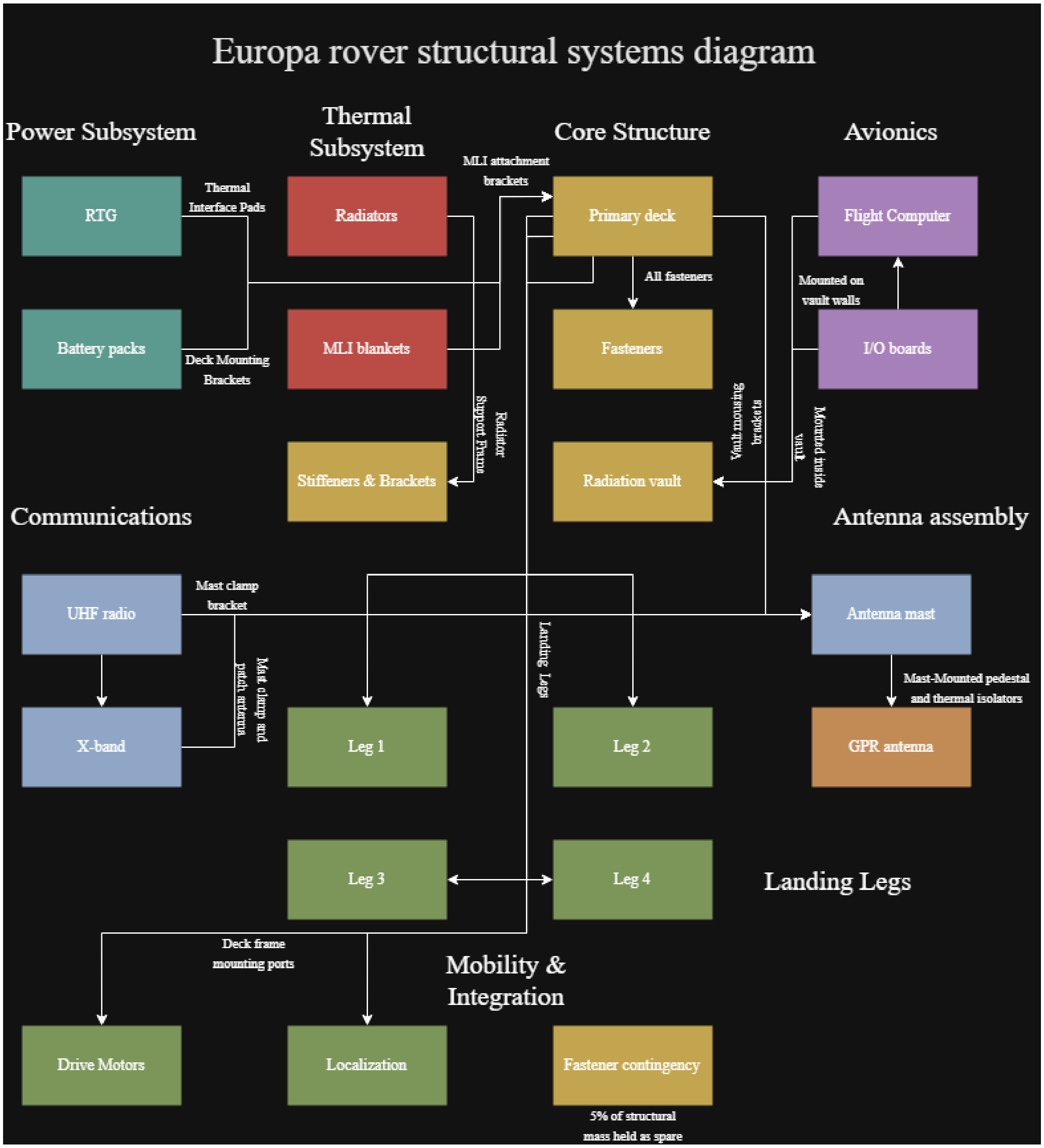



Figure 3: Rover systems diagram: subsystem-level interfaces across Power, Thermal, Core Structure, Avionics, Communications, Antenna Assembly, Landing Legs, and Mobility & Integration.

## 7.3 Structures and Mechanisms

### 7.3.1 Structural Requirements

- **REQ.03.02**: Antenna boresight within ±5° of nadir during sounding; rover halts traversal above 20° surface slope. (Structures, Science/Payload, Mobility)
- **REQ.07.01**: The rover chassis and landing leg structure shall accommodate vertical landing loads of 12$g$ and lateral loads of 6$g$ touchdown, with each leg sized to distribute peak contact

forces across four footpads without permanent deformation or buckling. (Structures, Mobility, GNC)

- **REQ.07.02**: Primary equipment deck, fasteners, and all structural material shall remain within specification (yield strength, fatigue life) under sustained 90 K cryogenic operation over the 30-sol mission lifetime. (Structures, Thermal, Power)
- **REQ.07.03**: The radiation shielding vault enclosing the flight computer and power distribution unit shall attenuate the Europa radiation environment to $\leq 1$ krad/sol at the electronics level, maintaining structural stiffness and mass allocation within the structures budget. (Structures, Power, Electronics, CDH)

### 7.3.2 Structural Sketch

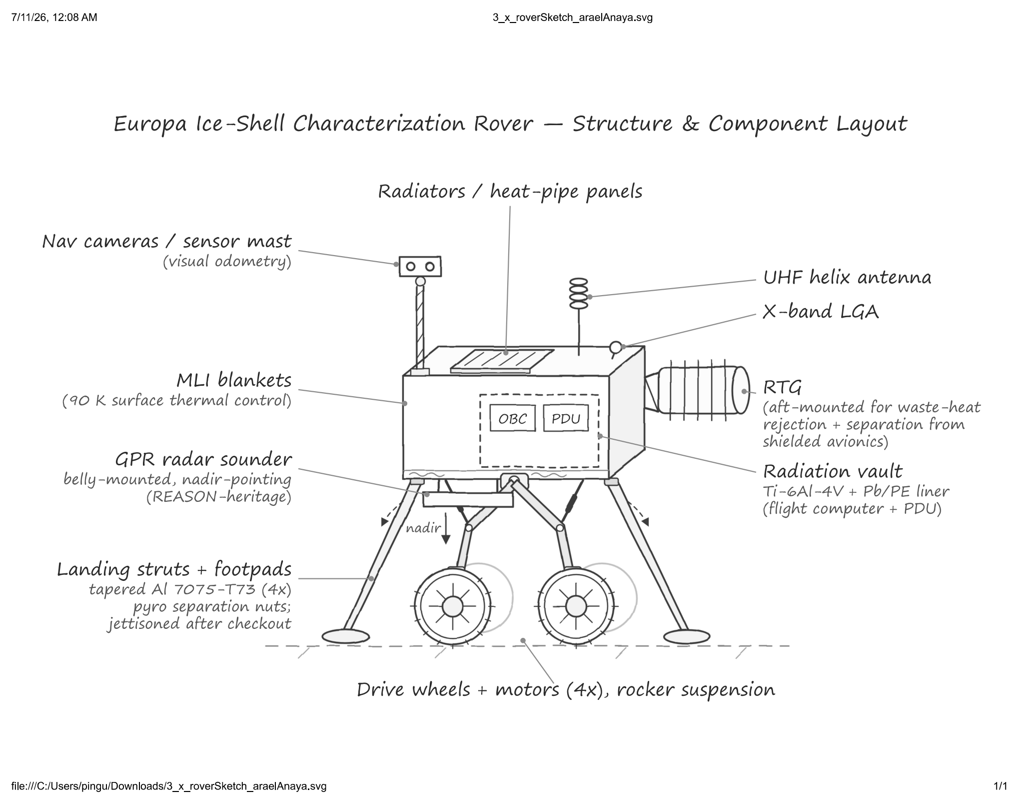



Figure 4: Annotated structural sketch of the rover chassis, deck, radiation vault, antenna mast, and landing-leg arrangement.

### 7.3.3 Mass Budget

Table 12 reproduces the mass budget in full, including the Mobility line item (Section 7.7, 3.6 Mobility Solution) and the REASON-heritage GPR sounder's own instrument mass (transceiver, digitizer, and antenna elements; the antenna mast/mounting pedestal is carried separately under Structure) and the GNC sensor suite (stereo nav cameras, MEMS IMU, fine sun sensor pair) that Section 7.7.4 costs and Table 15 powers but earlier revisions of this budget omitted as a line item. Dry mass totals 146.65 kg (129.55 kg core bus, including the 10.00 kg sounder, plus

17.10 kg of mobility hardware) against the 147.25 kg contingency cap, leaving 0.60 kg (0.4%) of margin. This cap is itself the contingency-inclusive figure, not a nominal allocation with contingency layered on top of it, so the 0.60 kg remaining is genuine design margin rather than double-counted reserve, though it is now thin enough that any further growth needs an explicit descope. If the REASON-heritage sounder's mass grows during CDR-level design, the first descope candidate is the 5.50 kg MEDA-class environment suite (Table 12): by Section 5's own account it supports interpretation of the sounder's returns but drives no Section 5 requirement on its own, making it the lowest-cost item to cut against the margin available.

| Subsystem | Component | Qty | Mass (kg) | Notes |
|---|---|---|---|---|
| Structure | Primary equipment decks/plates | 2 | 5.24 | Parametric sizing for avionics + GPR antenna footprint |
| Structure | Landing legs (struts + footpads) | 4 | 10.36 | 2.59 kg/leg, tolerates $< 12g$ vertical / $6g$ lateral touchdown |
| Structure | Stiffeners/brackets/frames | 1 | 1.88 | Prevents deck deflection under landing loads |
| Structure | Fasteners & support hardware | 1 | 1.42 | Cryogenic-rated titanium M5/M6 |
| Structure | Fastener contingency reserve | 1 | 0.73 | 5% structural mass contingency |
| Structure | Antenna mast & mounting bracket | 1 | 2.14 | GPR pedestal, Al tube + thermal isolators for nadir pointing |
| Structure | Radiation shielding vault enclosure | 2 | 2.08 | Ti-6Al-4V; attenuates dose per REQ.01.02 |
| Structure | Lead/polyethylene radiation liner | 1 | 1.56 | Selective shielding for flight computer |
| Structure | Thermal interface pads | 2 | 0.34 | Minimizes thermal short through vault wall at 90 K |
| Thermal | MLI attachment structure | 1 | 1.44 | Brackets/clips for MLI retention + radiator support |
| Thermal | Radiator support frame | 2 | 1.52 | Al tube frame, thermal decoupling |
| Thermal | MLI blankets | 1 | 6.32 | 0.35 kg/m$^2$, 18.05 m$^2$ equivalent coverage |
| Thermal | Radiators / heat-pipe panels | 2 | 9.58 | 4.79 kg/panel @ 6.0 kg/m$^2$, sized for 400–500 W rejection |
| Power | Battery packs (12-hr survival) | 1 | 19.2 | 150 Wh/kg $\times$ 16 kg +20% margin |
| Power | PDU + high-current cabling | 1 | 4.68 | Heritage EPS scaling |
| Power | RTG | 1 | 28.0 | Main power source |
| Power | Heater | 1 | 2.86 | Heating unit |
| Comms | UHF radio + cabling | 2 | 1.83 | Heritage UHF radios, 1.5–2 kg each |
| Comms | X-band transponder | 1 | 3.02 | DSN-compatible SDST heritage |
| Comms | Antennas (UHF helix / X-band LGA) | 1 | 0.86 | Short RF cabling |
| Avionics | Flight computer | 1 | 2.54 | OBC board + enclosure |
| Avionics | I/O / storage / interface boards | 1 | 1.52 | Logger + ADCs, includes enclosure |
| GNC | Stereo nav camera pair, MEMS IMU, fine sun sensor pair (aggregate) | 1 | 1.75 | Powered per the GNC rows of Table 15; fine sun sensor pair alone is 0.12 kg (Section 7.7.4 trade study) |
| Payload | GPR sounder electronics (REASON-heritage transceiver, digitizer, antenna elements) | 1 | 10.00 | Rover-adapted from flight REASON; mast/pedestal (2.14 kg) carried under Structure |
| Payload | MEDA-class environment suite | 1 | 5.5 | Perseverance MEDA heritage, $\sim$ 17 W op. power |
| Software | Flight / GNC / FDIR | 0 | 0 | No physical mass; impacts OBC/EPS sizing only |
| Harness & Integration | System harness / connectors / strain relief | 1 | 3.18 | 2–3% of dry mass |
| Mobility | Wheels, rocker suspension, drive motors, mast pan/tilt actuator (aggregate) | 1 | 17.10 | Per 3.6 Mobility Solution; not yet broken into individual component line items in the source spreadsheet |
| | | | **Dry Mass Total** | **146.65 kg** |
| | | | **Remaining Margin vs. Cap** | **0.60 kg (0.4%)** |
| | | | **Mass Cap Limit** | **147.25 kg** |

Table 12: Rover mass budget by subsystem and component.

| Subsystem | Total Mass (kg) | % Dry Mass |
|---|---|---|
| Structure | 25.75 | 17.56 |
| Thermal | 18.86 | 12.86 |
| Power | 54.74 | 37.33 |
| Comms | 5.71 | 3.89 |
| Avionics/OBC | 4.06 | 2.77 |
| GNC | 1.75 | 1.19 |
| Payload | 15.50 | 10.57 |
| Harness and Integration | 3.18 | 2.17 |
| Software | 0.00 | 0.00 |
| Mobility | 17.10 | 11.66 |
| **Total** | **146.65** | **100.00** |

Table 13: Mass budget rollup by subsystem. Payload includes both the 10.00 kg GPR sounder electronics and the 5.50 kg MEDA-class environment suite.

The Mobility line item's 17.1 kg (Section 7.7: wheels, rocker suspension, drive motors, mast pan/tilt actuator) and the corresponding 3.5 W mast pan/tilt actuator draw (Table 15, Section 7.4, during the 1-hour Traverse Planning mode) are both carried as explicit rows above.

### 7.3.4 Structures Trade Study

**Study:** Analysis of Landing Leg Structure. Options were scored against six pairwise-weighted criteria (Load Bearing Capability 0.25, Mass per Leg 0.20, Safety Margin at Yield 0.25, Thermal Conductivity 0.18, Manufacturability & TRL 0.25, Cryogenic Ductility 0.05).

| Criterion (weight) | Tapered Strut Al 7075-T73 | Cross-Braced Tube + Diagonals | Ti-6Al-4V Folding Hinge (Hybrid) |
|---|---|---|---|
| Load Bearing (0.25) | 1.25 | 1.25 | 1.00 |
| Mass per Leg (0.20) | 0.80 | 0.60 | 0.60 |
| Safety Margin at Yield (0.25) | 1.25 | 1.25 | 1.00 |
| Thermal Conductivity (0.18) | 0.54 | 0.36 | 0.72 |
| Manufacturability & TRL (0.25) | 1.25 | 0.75 | 0.50 |
| Cryogenic Ductility (0.05) | 0.20 | 0.25 | 0.15 |
| **Total Score** | **5.29** | 4.46 | 3.97 |

Table 14: Landing leg structure trade study.

**Winning option: Baseline tapered-strut Al 7075-T73 leg.** It matches the top score on Load Bearing Capability and Safety Margin at Yield (both 5/5) while separating itself on Manufacturability & TRL (5/5, off-the-shelf/heritage design, simple machining), which is the largest single differentiator against the cross-braced and folding-hinge alternatives. This is consistent with the 2.59 kg/leg figure and material call-out already carried in the mass budget (Table 12) and the tapered-strut geometry drawn in Figure 4.

## 7.4 Electrical Power System

### 7.4.1 Power Requirements

- **REQ.05.03**: Per-sol energy draw across all subsystems shall not exceed 90% of the RTG's nominal daily energy output, reserving $\geq$ 10% margin for anomaly response, heater activations, and unplanned relay passes. (Power, CDH, Thermal, Mobility)

- **REQ.08.01**: The power system shall generate $\geq 110$ We (BOL) via an MMRTG-class RTG and deliver $\geq 80$ We continuous upon Europa arrival, providing $\geq 1.9$ kWh usable energy per sol across the 30-sol surface mission. (Power, Thermal, System)
- **REQ.08.02**: Energy storage shall provide $\geq 2.4$ kWh usable capacity to (1) load-level peak draws (drive motors, radar transmission, X-band downlink) beyond continuous RTG output, and (2) sustain night-mode survival heating and critical avionics for $\geq 12$ hours, maintaining cell temperature above $-20°$C per REQ.05.01. (Power, Thermal)
- **REQ.03.03**: Dedicated peak power allocation reserved for radar transmission pulses; drive motors commanded off during each active sounding window. (Science/Payload, Power, Mobility)
- **REQ.08.03**: The power distribution unit shall autonomously detect and isolate a faulted load bus within 100 ms of overcurrent/undervoltage detection, maintain uninterrupted power to critical loads (flight computer, survival heaters, UHF radio), and log each fault event to non-volatile memory, without ground intervention. (Power, CDH, Electronics)

### 7.4.2 Power Budget & Profile

Table 15 gives the per-component power budget; the RTG delivers a constant 110 W generated power baseline against which every mode is evaluated. Seven operating modes were simulated across a 24-hour sol: Traverse Planning (1 hr), Traverse (4 hr), Radar Sounding (5 hr), Comms Relay Pass (1.5 hr), Map Generation & Consensus (4.5 hr), Systems Warm-Up (the ramp-up sub-window at the tail of Night Low-Power as heaters and avionics come up ahead of Traverse Planning, per REQ.09.02's 60-minute cold-start bound), and Night Low-Power (8 hr total, inclusive of the Warm-Up ramp), summing to the full 24-hour cycle. Table 16 reports Warm-Up's average power separately from steady-state Night even though both sit inside that same 8-hour block.

| Subsystem | Component | Peak (W) | Avg (Wh/hr) | Duration (h) |
|---|---|---|---|---|
| Payload | GPR Radar Sounder | 45 | 225.0 | 5.0 |
| Payload | MEDA-class Environment Suite | 17 | 85.0 | 5.0 |
| EPS | Battery Packs | 0 | 0.0 | 24.0 |
| EPS | PDU (PCDU) | 12 | 288.0 | 24.0 |
| Mobility | Drive Motors (4x) | 84 | 336.0 | 4.0 |
| GNC | Flight Computer (OBC) | 18 | 432.0 | 24.0 |
| GNC | I/O & Storage Boards | 6 | 96.0 | 16.0 |
| GNC | Nav Cameras | 4 | 20.0 | 5.0 |
| Thermal | Heaters | 40 | 320.0 | 8.0 |
| Thermal | Thermal Sensors | 0.5 | 12.0 | 24.0 |
| Comms | UHF Radio | 35 | 210.0 | 6.0 |
| Comms | X-band Transponder | 45 | 67.5 | 1.5 |
| Mobility | Mast Pan/Tilt Actuator | 3.5 | 3.5 | 1.0 |
| **Total** | | | **2095.0 Wh/sol** | |

Table 15: Power budget by component (peak draw and average energy consumption per sol). The Mobility mast actuator (Section 7.7, 3.6 Mobility Solution) draws its 3.5 W during Traverse Planning, when the rover repositions the mast/antenna assembly ahead of the next drive arc. The MEDA-class environment suite (Section 5) is duty-cycled inside the same 5-hour Radar Sounding window as the GPR sounder, logging environmental context alongside each radar trace, and is carried within the payload power allocation rather than as a continuous draw.

| **Mode** | Traverse | Sounding | Comms Relay | Map Gen. | Traverse Plan. | Warm-Up | Night |
|---|---|---|---|---|---|---|---|
| Avg. Power (W) | 124.5 | 98.5 | 116.5 | 71.5 | 44.0 | 121.5 | 70.5 |

Table 16: Average power draw by operating mode. Sounding sums the GPR sounder, the duty-cycled PDU/OBC/I/O baseline, thermal sensors, and the MEDA-class environment suite, which shares the same 5-hour window (Table 15).

The battery (Max Charge 2400 Wh, Min Safe Charge 480 Wh) never approaches its safety floor across the simulated 24-hour sol: lowest observed charge is 2342.1 Wh (only $\sim$ 58 Wh, or 2.4%, below full), and highest charge/discharge rates are 69.5 W/14.5 W respectively. Total energy consumption is 2095.0 Wh/sol (Table 15, inclusive of the MEDA-class suite's 85 Wh over its 5-hour duty window), an average power draw of 87.3 W against 110 W generated. This is consistent with REQ.05.03's 90%-of-RTG-output cap with 10% reserve: $87.3/110 = 79.4\%$, inside the cap with 10.6 percentage points to spare. Drive motors and the radar sounder are never scheduled concurrently in the mode matrix (REQ.03.03's stop-and-sound logic), and the flight computer, PDU, and thermal sensors stay powered continuously across every mode as the "always-on" core avionics load.

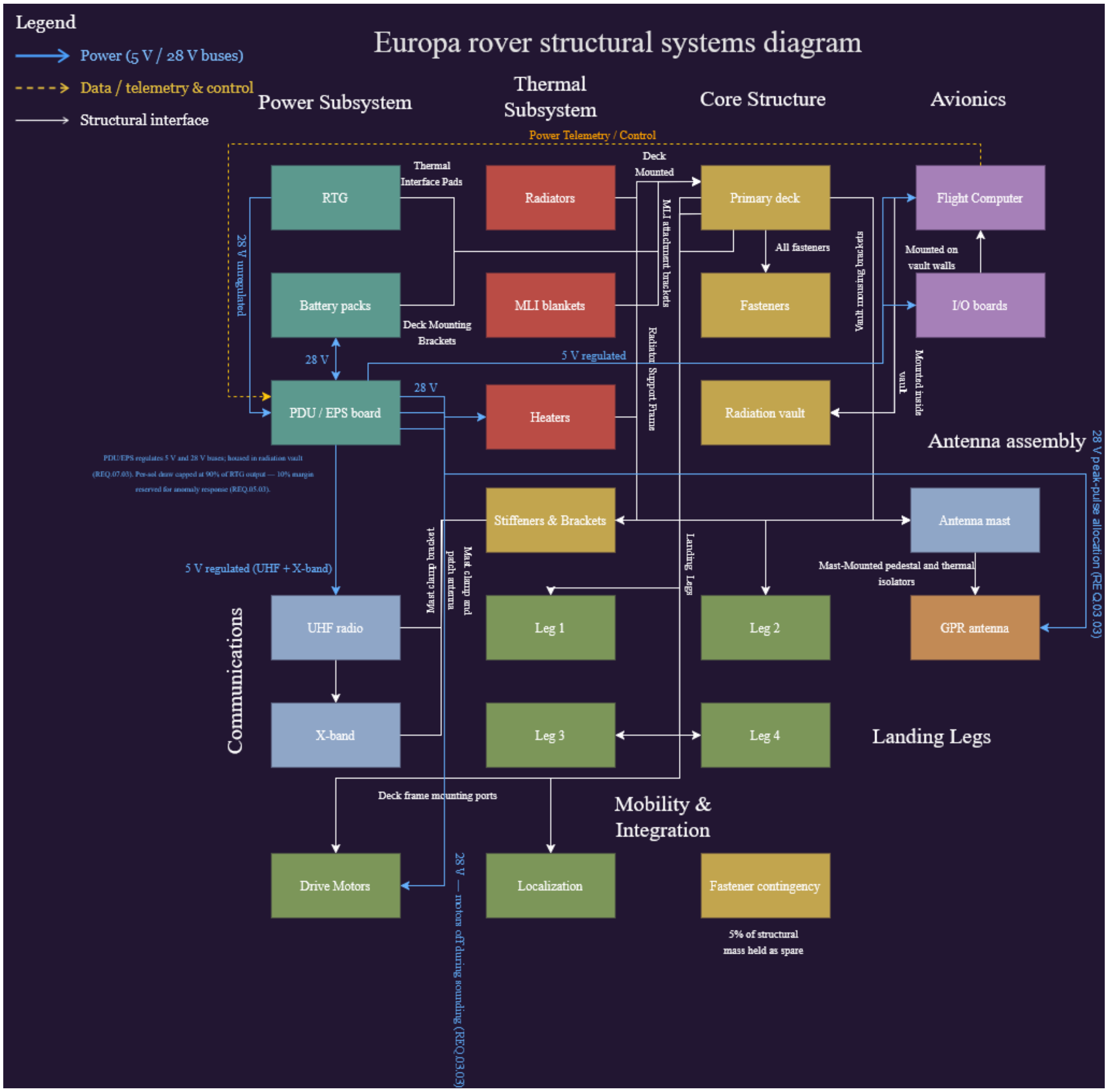



Figure 5: Power systems diagram: RTG and battery packs feeding the PDU/EPS board over regulated 28 V/5 V buses, with per-sol draw capped at 90% of RTG output (REQ.05.03) and peak-pulse allocation reserved for the radar sounder (REQ.03.03).

### 7.4.3 Power Trade Study

**Study:** Analysis of Power Source. Options were scored against five pairwise-weighted criteria (Mass 0.13, Continuous Power 0.28, Complexity/Operations 0.13, Radiation & Cryo Robustness 0.25, Availability/TRL 0.23).

| **Criterion (weight)** | **Solar Arrays + Li-ion** | **RTG** | **$H_2/O_2$ Fuel Cells** |
|---|---|---|---|
| Mass (0.13) | 0.25 | 0.38 | 0.13 |
| Continuous Power (0.28) | 0.28 | 1.38 | 0.55 |
| Complexity/Operations (0.13) | 0.25 | 0.63 | 0.25 |
| Radiation & Cryo Robustness (0.25) | 0.50 | 1.25 | 0.75 |
| Availability/TRL (0.23) | 0.68 | 1.13 | 0.68 |
| **Total Score** | 1.95 | **4.75** | 2.35 |

Table 17: Power source trade study.

**Winning option: RTG (radioisotope thermoelectric generator).** At 5.2 AU, solar irradiance has collapsed too far for arrays to deliver continuous power, and the RTG scores a perfect 5/5 on four of the five criteria (Continuous Power, Complexity/Operations, Radiation & Cryo Robustness, and Availability/TRL), losing only on raw mass (score 3/5, still the highest-weighted criterion it does not win). This selection is what sizes the 28 kg RTG line item in the mass budget (Table 12) and directly supports REQ.08.01's $\geq$ 110 We beginning-of-life requirement.

## 7.5 Thermal Management

### 7.5.1 Thermal Requirements

- **REQ.05.01**: Maintain avionics and battery cells within 0–40°C during active operations and above −20°C during night-mode low-power management, using RTG-waste-heat-powered active heater elements. (Thermal, Power, Structures)
- **REQ.09.01**: Reject up to 500 W of combined electronics dissipation and routed RTG waste heat through dedicated radiator panels, keeping the avionics vault below 40°C during peak daytime sounding/traverse operations. (Thermal, Electronics, Power, CDH)
- **REQ.09.02**: Preheat all drive-train actuators and deployment mechanisms above −40°C before any commanded motion, completing cold-start warm-up within 60 minutes of the 90 K surface baseline. (Thermal, Mobility, Structures)
- **REQ.09.03**: Combine MLI with routed RTG waste heat to bound the equipment-enclosure heat leak such that supplemental electric survival heating does not exceed 50 W and stays within the REQ.05.03 per-sol energy budget. (Thermal, Power, Structures)

### 7.5.2 Heat Load Budget

Table 18 lists per-component power usage, efficiency, and resulting heat generation. Aggregate power usage across all listed components is 306.5 W at a blended efficiency of 0.942, for a total heat generation of 17.76 W (this is the electronics dissipation term; it excludes the RTG's own $\sim$ 2 kW continuous thermal rejection, which is routed separately as the primary survival heat source per REQ.09.03).

| Board | Component | Power (W) | Eff. | Heat Gen. (W) |
|---|---|---|---|---|
| Payload | GPR Radar Sounder (REASON-heritage) | 45 | 0.95 | 2.25 |
| Payload | MEDA-class Environment Suite | 17 | 0.95 | 0.85 |
| EPS | Battery Packs | 0 | 1.00 | 0.00 |
| EPS | PDU (PCDU) | 12 | 0.93 | 0.84 |
| Mobility | Drive Motors (4x) | 84 | 0.90 | 8.40 |
| GNC | Flight Computer (OBC) | 18 | 0.95 | 0.90 |
| GNC | I/O & Storage Boards | 6 | 0.95 | 0.30 |
| GNC | Nav Cameras (stereo VO) | 4 | 0.95 | 0.20 |
| Thermal | Heaters | 40 | 1.00 | 0.00 |
| Thermal | Thermal Sensors | 0.5 | 0.95 | 0.025 |
| Comms | UHF Radio (Electra-Lite heritage) | 35 | 0.95 | 1.75 |
| Comms | X-band Transponder | 45 | 0.95 | 2.25 |
| **Total** | | **306.5** | **0.942** | **17.76** |

Table 18: Heat load budget by component. The MEDA-class environment suite (Section 5) is duty-cycled inside the same 5-hour Radar Sounding window as the GPR sounder rather than drawing continuously.

In addition to the equipment heat load above, a point-mass thermal equilibrium analysis was performed for the avionics enclosure (area 0.12 m$^2$, $\alpha = 0.20$, $\varepsilon = 0.85$). The hot case ($Q_{int} = 20$ W, daytime operations) equilibrates at $T_{hot} \approx 248$ K ($-25$°C); the cold case ($Q_{int} = 5$ W, night survival) equilibrates at $T_{cold} \approx 175$ K ($-98$°C): a $\sim 73$ K swing. Unlike a Mars-class mission, the problem on Europa is not rejecting heat but retaining it: solar irradiance at 5.2 AU collapses to $\sim 50$ W/m$^2$, so internal dissipation dominates the energy balance, and even the operating case falls below the $\sim 253$ K ($-20$°C) lower bound for Li-ion battery charging. The survival case sits far outside any electronics or battery limit and would freeze the system without intervention, which is exactly why MLI, routed RTG waste heat, and controlled survival heaters (REQ.09.03) are required rather than optional.

Closing the gap between the 248 K hot-case equilibrium and REQ.05.01's 273 K (0°C) active-ops floor is a heater-sizing problem, not an open item. Using the same radiative balance ($Q \approx \varepsilon\sigma A T^4$) that produced the 248 K point, reaching 273 K requires raising the heat delivered to the enclosure to $\varepsilon\sigma A(273\,\mathrm{K})^4 \approx 32.1$ W. Naively subtracting the 20 W hot-case dissipation would suggest a 12.1 W makeup, but the 248 K equilibrium already carries a small absorbed-solar contribution ($\alpha SA$, on the order of 1–2 W at Europa's $\sim 50$ W/m$^2$ irradiance and $\alpha = 0.20$) folded into that balance, so the true supplemental heater makeup is closer to 10 W. RTG waste heat ($\sim 2$ kW thermal at the source, Section 7.5.3) is routed to the avionics enclosure specifically to cover this gap; even treating the full 10 W as if it were drawn instead from the supplemental electric survival heaters, it consumes only about a fifth of REQ.09.03's 50 W cap, leaving $\sim 40$ W of headroom for the colder night-survival case. The active-ops thermal floor therefore closes with margin using hardware already in the design (MLI, routed RTG waste heat, and the survival heaters sized below). The night case closes the same way: holding the 253 K ($-20$°C) Li-ion charging floor at the 5 W night-survival internal dissipation requires the enclosure to receive $\varepsilon\sigma A(253\,\mathrm{K})^4 \approx 23.7$ W total, or about 19 W of supplemental heat once the 5 W of internal dissipation already present is credited, comfortably inside REQ.09.03's 50 W cap with $\sim 31$ W to spare.

### 7.5.3 Thermal Trade Study

**Study:** Analysis of Night Survival Thermal Architecture. Options were scored 1–5 against five pairwise-weighted criteria (Mass 0.13, Power Demand 0.25, Complexity/Integration 0.13, Radiation Durability 0.25, Reliability/TRL 0.25).

| Criterion (weight) | MLI + Heaters | Aerogel + Heaters | PCM + Heat Pipes |
|---|---|---|---|
| Mass (0.13) | 0.52 | 0.52 | 0.26 |
| Power Demand (0.25) | 1.25 | 0.75 | 1.00 |
| Complexity/Integration (0.13) | 0.52 | 0.39 | 0.26 |
| Radiation Durability (0.25) | 1.25 | 0.75 | 0.75 |
| Reliability/TRL (0.25) | 1.25 | 0.75 | 0.50 |
| **Total Score** | **4.79** | 3.16 | 2.77 |

Table 19: Night-survival thermal architecture trade study.

**Winning option: MLI and heaters** (4.79 vs. 3.16 for aerogel + heaters and 2.77 for PCM with heat pipes). RTG waste heat ($\sim$ 2 kW thermal) serves as the primary survival heat source; MLI limits radiative loss so that supplemental electric survival heating stays at or below 50 W (REQ.09.03), keeping night thermal draw inside the per-sol energy budget (REQ.05.03).

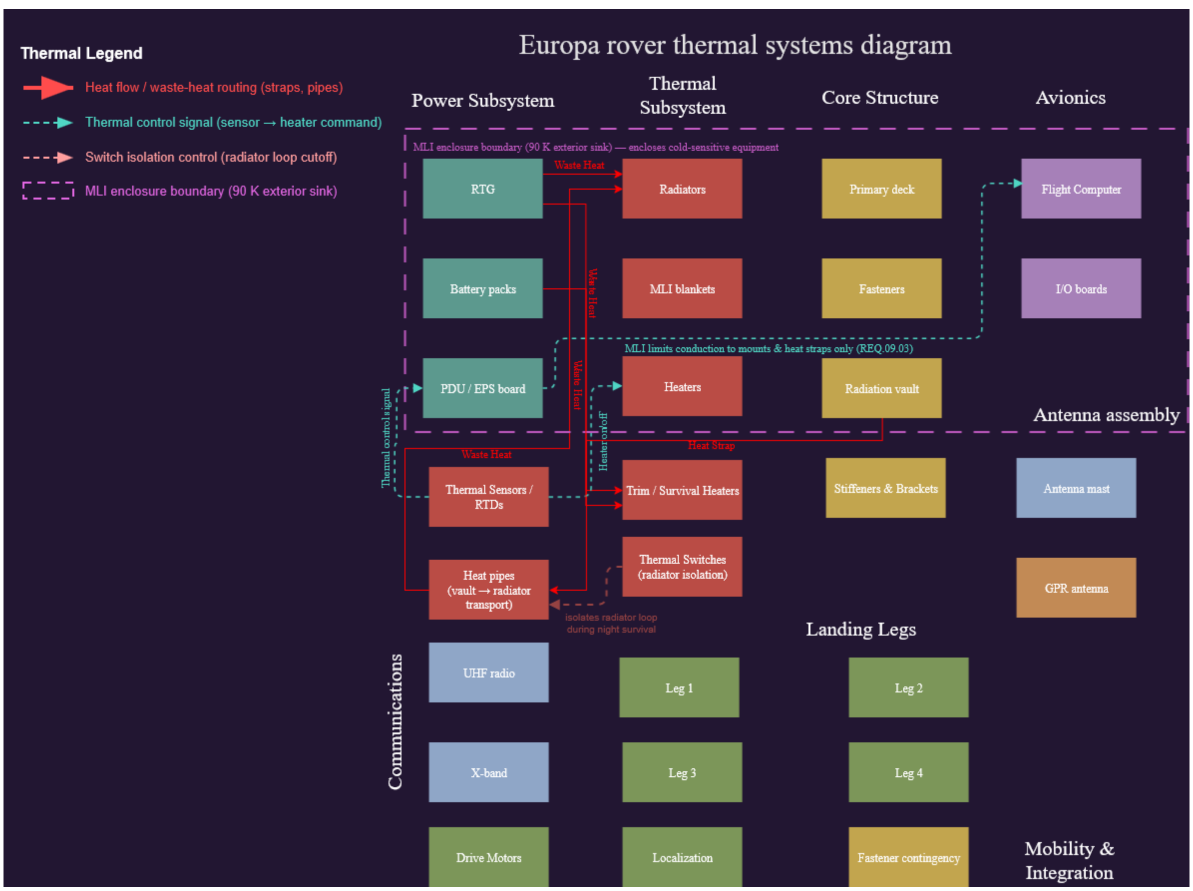



Figure 6: Thermal systems diagram: RTG/heater waste-heat routing to radiators, MLI enclosure boundary, and radiator-loop isolation switches for night survival.

## 7.6 Communications

### 7.6.1 Communications Requirements

- **REQ.01.03**: Maintain $\geq 2$ active relay nodes at all times during surface operations. (System, COMMs)
- **REQ.04.01**: Inter-agent mesh network shall achieve terrain-map consensus within $\leq 10$ minutes under $\leq 30\%$ packet loss on any single link. (COMMs, CDH)
- **REQ.10.01**: Deliver $\geq 55$ MB/Earth-day compressed data at $\geq 128$ kbps return / $\geq 8$ kbps forward. (COMMs, CDH, Science/Payload, Power)
- **REQ.10.02**: Obtain $\geq 2$ usable relay contacts/sol totaling $\geq 90$ min above a 10° elevation mask; end-to-end latency $\leq 26$ hours. (COMMs, CDH, Science/Payload, GNC)
- **REQ.10.03**: Close the proximity link with $\geq 3$ dB margin at the worst-case 588 km slant range; retain an X-band DTE beacon $\geq 5$ bps as a relay-loss fallback. (COMMs, Power, Structures, CDH)

### 7.6.2 Comm Architecture & Assumptions

The rover's next communication node is the nearer of the two orbital relay agents; the mission does not route science data through a direct link to Earth in nominal operations. The link uses a UHF proximity band on the CCSDS Proximity-1 return/forward pair [9]: 401.6 MHz return, 437.1 MHz forward, the same split flown by Electra and Electra-Lite on the Mars Relay Network [11]. Desired margin is 3 dB above the $E_b/N_0$ required for a post-decoding bit error rate of $10^{-6}$ (CCSDS turbo coding, rate 1/2, threshold near 1.5 dB [10], so the link must clear 4.5 dB total): a conservative cushion given that the relay's exact orbit geometry, the rover's helix gain pattern, and Jupiter's synchrotron noise contribution are all still modeled rather than measured. Worst-case geometry assumes both relays fly a 200 km circular Europa orbit; with a 10° elevation mask, the slant range at the edge of a pass is 588 km, and every link budget term uses that worst case.

Rover transmit power is 5 W RF into a 5.0 dBi quadrifilar helix (near-hemispherical pattern, so the rover can close a pass without pointing at anything); the relay carries 8 W RF into a 7.0 dBi patch array, since its power budget has more slack. System noise temperature is 3000 K at the relay receiver and 4000 K at the rover receiver: the number that most separates a Europa proximity link from a Mars one, since at 400 MHz the receiver picks up galactic background noise on top of Jupiter's synchrotron emission. The return link runs 128 kbps (sized to clear 55 MB in the 90-minute relay window) and the forward link runs 8 kbps (commands only). Figure 7 shows the full data-flow architecture from the payload through CDH, the modem/transceiver chain, the RF front end, and out to the relay agents and, ultimately, the DSN [12].

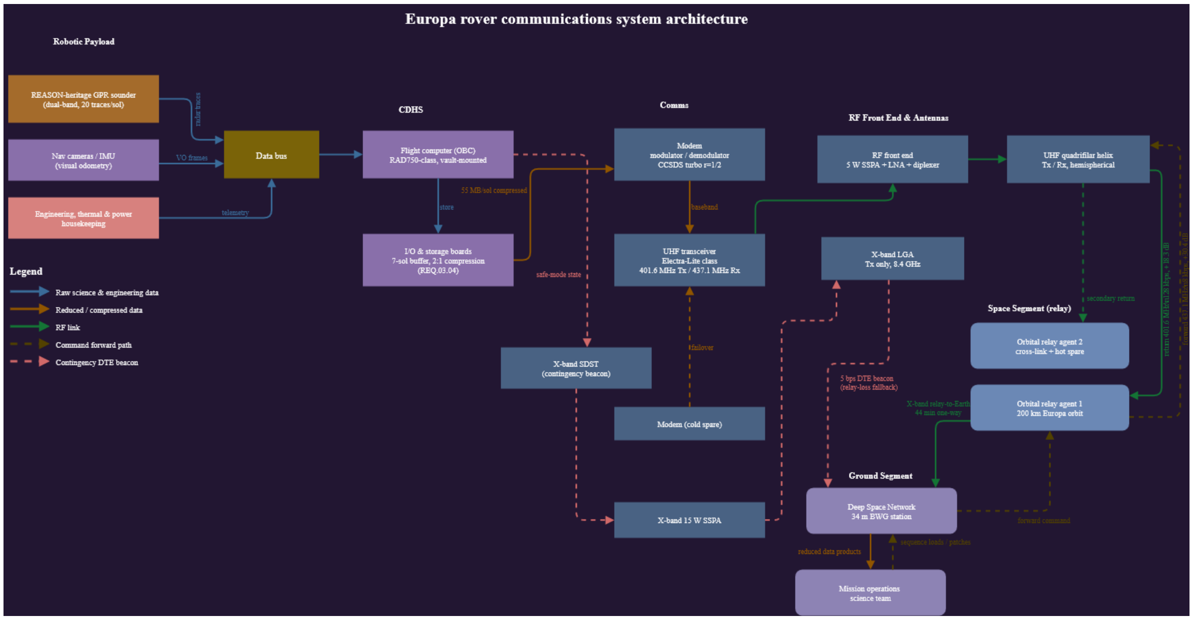



Figure 7: Communications system architecture: data flow from payload/nav/health sources through CDH, comms, and the RF front end to the orbital relay agents and DSN ground segment.

### 7.6.3 Partial Link Budget

Both directions close with wide margin, following the link-budget convention in the DSN Telecommunications Link Design Handbook [8]: 18.3 dB on the return (uplink, rover→relay) and 30.4 dB on the forward (downlink, relay→rover), each measured over the raw 1.5 dB decoding threshold. Measured instead against the full 4.5 dB requirement (the 1.5 dB decoding threshold plus the 3 dB desired cushion, Section 7.6.2), that is 15.3 dB of margin on the return and 27.4 dB on the forward. Table 20 breaks the return link down term by term, reporting the 18.3 dB figure as margin over the 1.5 dB threshold.

| Term | Value (dB) |
|---|---:|
| $P_t$ (5 W) | +6.99 |
| $G_t$ (rover helix) | +5.00 |
| $G_r$ (relay patch array) | +7.00 |
| $L_a$ (polarization + plasma) | −0.51 |
| $L_l$ (line loss) | −1.49 |
| $\lambda^2$ | −2.54 |
| $(4\pi R)^2$ (space loss @ 588 km) | −137.37 |
| $k$ | +228.60 |
| $T_s$ (3000 K) | −34.77 |
| $R_b$ (128 kbps) | −51.07 |
| $E_b/N_0$ | **+19.84** |
| Margin over 1.5 dB requirement | **+18.3** |

Table 20: Return-link (rover-to-relay) budget, term by term.

As a direct-to-Earth sanity check: the rover's X-band low-gain antenna against a DSN 34 m station closes at roughly 6.9 bps at opposition (4.2 AU) and 3.2 bps at conjunction (6.2 AU) while holding the same 4.5 dB margin: enough for a state-of-health beacon, but a single 55 MB sol of science data would take over two years to clear at that rate. This is why the relay architecture is foundational rather than a convenience (REQ.10.03's 5 bps fallback floor is set directly from this analysis).

### 7.6.4 Communications Subsystem Diagram

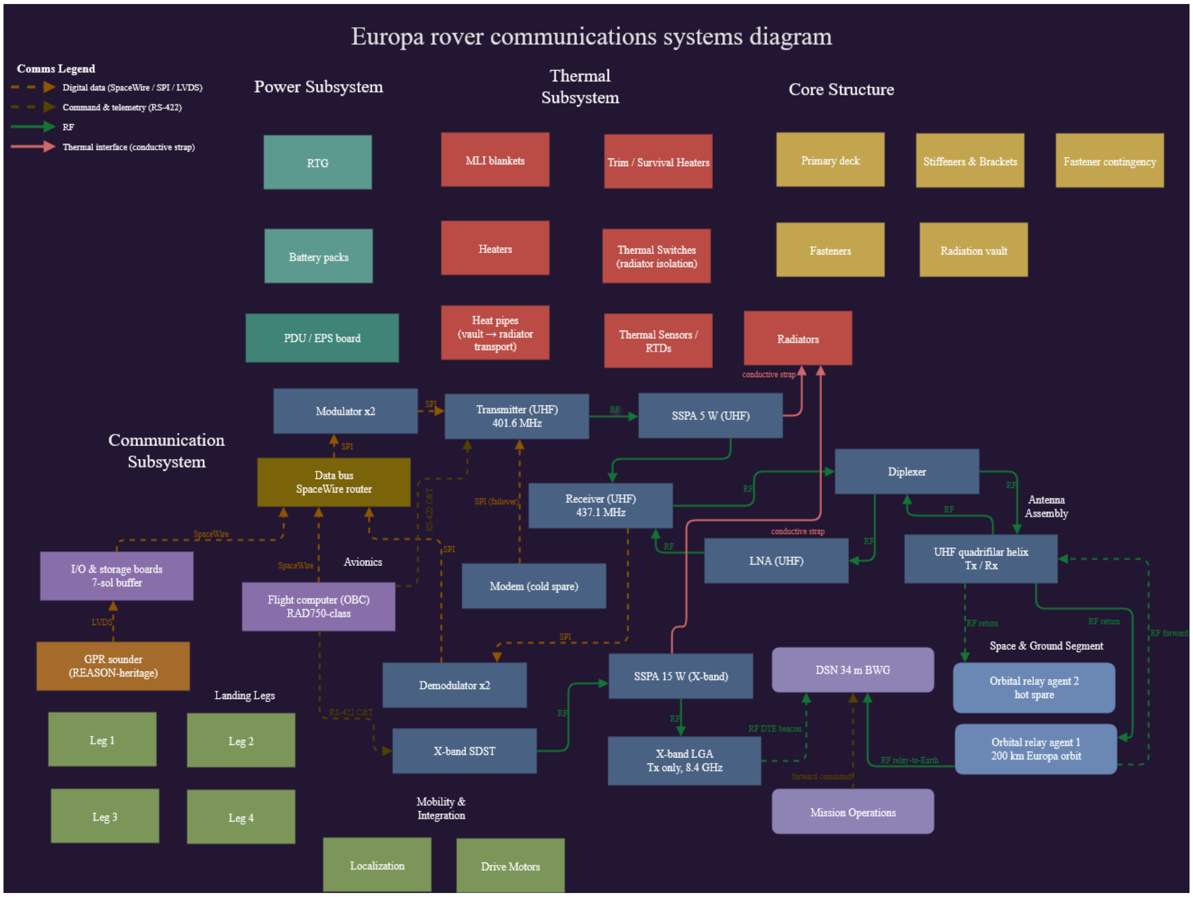



Figure 8: Communications systems diagram: modulator/transceiver chain, RF front end, and antenna assembly, with digital-data, command/telemetry, RF, and thermal-interface signal paths distinguished.

### 7.6.5 Communications Trade Study

**Study:** Analysis of Rover-to-Relay Proximity Link Configuration. Options were scored against five pairwise-weighted criteria (Mass 0.13, Power Demand 0.25, Data Rate Capability 0.25, Pointing & Ops Complexity 0.13, Reliability/TRL 0.25).

| Criterion (weight) | UHF Helix | S-band Patch | X-band Gimballed HGA |
|---|---|---|---|
| Mass (0.13) | 0.65 | 0.52 | 0.26 |
| Power Demand (0.25) | 1.00 | 0.75 | 0.50 |
| Data Rate Capability (0.25) | 1.00 | 1.25 | 1.25 |
| Pointing & Ops Complexity (0.13) | 0.65 | 0.52 | 0.13 |
| Reliability/TRL (0.25) | 1.25 | 0.75 | 0.50 |
| **Total Score** | **4.55** | 3.79 | 2.64 |

Table 21: Rover-to-relay proximity link configuration trade study.

**Winning option: UHF quadrifilar helix** (401.6/437.1 MHz). The X-band option has the raw throughput but fails the other criteria: a two-axis gimbal is a moving mechanism that has to survive 90 K and 100 krad, and it needs closed-loop tracking of a relay the rover cannot see coming. S-band buys data rate the mission does not need, since the UHF return already closes at 128 kbps with 18.3 dB of margin and is bounded by the transceiver's 2048 kbps ceiling rather than by RF. The helix is already in the mass budget at 0.86 kg and carries TRL 9 through Electra and Electra-Lite heritage on exactly this kind of proximity link.

## 7.7 GNC and Mobility

### 7.7.1 GNC Requirements

- **REQ.11.01**: Determine attitude to $\pm 0.5°$ ($3\sigma$) roll/pitch and heading to $\pm 0.25°$ ($3\sigma$) at every sounding stop, propagating inertially between arcs $\leq 5$ m and taking an absolute heading fix from fine sun sensors at each arc end. (GNC, Mobility, Science/Payload, CDH)
- **REQ.11.02**: Assess terrain $\geq 8$ m ahead via stereo nav cameras before each drive arc; autonomously reject arcs with a step $> 0.20$ m, slope $> 20°$, or roughness $> 0.7$; stop within 0.25 m of the assessed safe limit. (GNC, Mobility, Science/Payload, CDH)
- **REQ.11.03**: Follow the planned path with cross-track deviation $\leq 0.5$ m over a 50 m segment and stop within $\pm 0.5$ m of each commanded waypoint, using differential drive with wheel slip estimated from the odometry/visual-odometry residual. (GNC, Mobility, Science/Payload)
- **REQ.11.04**: Halt and enter a navigation hold whenever $1\sigma$ position uncertainty exceeds 0.5 m; re-initialize via two-way ranging to $\geq 2$ relay agents plus local/orbital terrain correlation; do not resume until uncertainty is restored below 0.5 m. (GNC, COMMs, CDH, Mobility)

### 7.7.2 Mobility Solution Sketch

**Configuration: Four-Wheel Differential Drive on a Single-Pivot Rocker.** Four independently driven wheels ride on a single-pivot rocker, with no steering actuators. The chassis is the 0.90 m × 0.60 m equipment deck already carried in the mass budget, held 0.35 m above the surface so the belly-mounted sounder keeps its standoff. Wheelbase is 0.70 m and track is 0.80 m; wheels are 0.32 m in diameter, 0.20 m wide, with a rigid aluminum rim carrying 24 hardened cleats [15]. Commanded speed is 2 cm/s, driven in arcs no longer than 5 m.

Four hours of drive time at 2 cm/s gives a raw 288 m per sol; a 65% duty cycle after subtracting stops for imaging and hazard assessment (REQ.11.02's per-arc terrain check and the sun-sensor heading fix taken at each 5 m arc boundary, REQ.11.01) brings that to the 187 m/sol capability called out in Figure 9, still nearly double REQ.06.01's 100 m/sol floor.

Wheels were chosen over hopping or legged locomotion because the mission needs a spatially continuous subsurface profile: REQ.03.01 calls for a radar trace every 100 m or better along a single transect, and every landing event in a hop or step creates a gap where the localization solution has to start over. A hop that lands wrong on a 90 K ice block can end the mission on the spot, with no window for Earth to intervene given the 44-minute one-way delay; a wheeled rocker instead degrades slowly, bearing by bearing, over many sols: a failure mode better suited to a mission with no recovery crew.

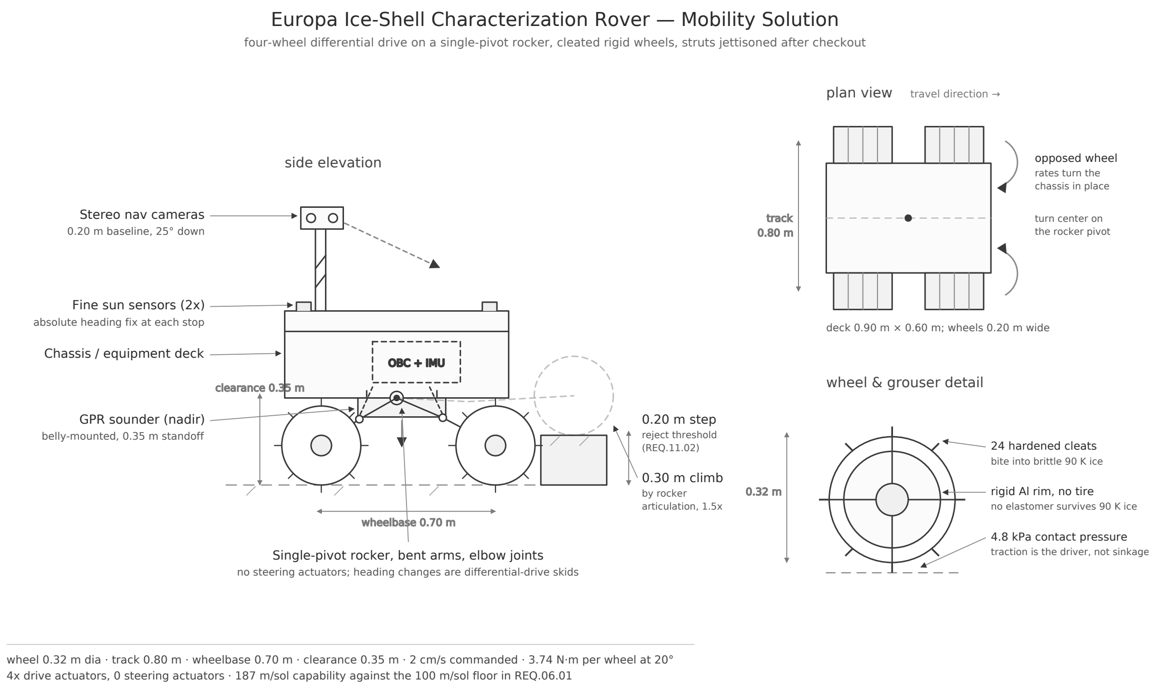



Figure 9: Mobility solution: side elevation, plan view, and wheel detail.

Steering is handled entirely by differential drive: opposing wheel rates rotate the chassis about the rocker pivot, so every heading change is a deliberate skid across the ice. Skid steering slips badly on hard ice ($\mu \approx 0.5$ [13]), so wheel odometry is dropped as a position source and used instead as a slip observable: the residual between encoder counts and stereo visual odometry becomes the slip estimate, the same approach the MER rovers used on Martian sand [14]. Heading is re-anchored with an absolute sun-sensor fix at the end of every 5 m arc so error never integrates across a full traverse segment.

### 7.7.3 Localization and Pointing Budget

Table 22 closes the 50 m segment localization error budget: root-sum-square of 0.46 m against the 0.5 m limit in REQ.02.01, an 8% margin.

| Error source over a 50 m segment | Contribution (m) |
|---|---|
| Visual odometry scale and feature residual | 0.30 |
| Heading knowledge (0.271°) | 0.24 |
| Wheel slip on cleated water ice | 0.20 |
| Inter-agent range residual | 0.15 |
| Root-sum-square | **0.46** |
| REQ.02.01 limit | 0.50 |

Table 22: Localization error budget, one 50 m traverse segment.

In addition to this position-error rollup, the source pointing budget defines per-component alignment requirements rather than a single combined pointing RSS: the GPR antenna boresight must hold ±5° of nadir (±0.5° stop-to-stop repeatability, REQ.03.02); the fine sun sensor pair resolves the Sun vector to ±0.1°, which folds into a ±0.25° heading solution after ephemeris and mounting alignment error (REQ.11.01); and the stereo nav camera boresight sits 25° below the local horizontal with a 0.20 m step-detection capability at 8 m range (REQ.11.02).

#### 7.7.4 GNC Trade Study

**Study:** Analysis of the absolute heading reference for surface navigation. Options were scored against five pairwise-weighted criteria (Mass 0.12, Power Demand 0.23, Complexity/Cost 0.12, Accuracy 0.30, Environmental Tolerance 0.23).

| Criterion (weight) | Fine Sun Sensor Pair | COTS Star Tracker | FOG Gyrocompassing |
|---|---|---|---|
| Mass (0.12) | 0.60 | 0.48 | 0.24 |
| Power Demand (0.23) | 1.15 | 0.69 | 0.46 |
| Complexity/Cost (0.12) | 0.48 | 0.24 | 0.12 |
| Accuracy (0.30) | 1.20 | 1.50 | 0.90 |
| Environmental Tolerance (0.23) | 1.15 | 0.46 | 0.46 |
| **Total Score** | **4.58** | 3.37 | 2.18 |

Table 23: Absolute heading reference trade study.

**Winning option: Fine sun sensor pair with onboard ephemeris.** The star tracker wins on raw accuracy and loses everywhere else: its detector degrades measurably under the Jovian dose, it needs a baffle against Jupiter (which subtends ∼ 11.8° from the surface and never sets on the sub-Jovian hemisphere), and it costs more than the rest of the GNC sensor suite combined. Gyrocompassing is the interesting failure: a 0.01°/h FOG can in principle find north off Europa's 4.22°/h rotation, but that signal is only 28% of what the same instrument would see on Earth, and the fiber coil and light source are not qualified at 90 K or 100 krad. The sun sensor is the cheap answer: at 5.2 AU the Sun delivers only 50 W/m$^2$, but its angular diameter collapses to 0.103°, so a photodiode sensor is looking at a sharper point source than it would at Earth. Two of them cost 0.12 kg and under 1 W, and Cassini flew the same class of sensor further out than Jupiter.

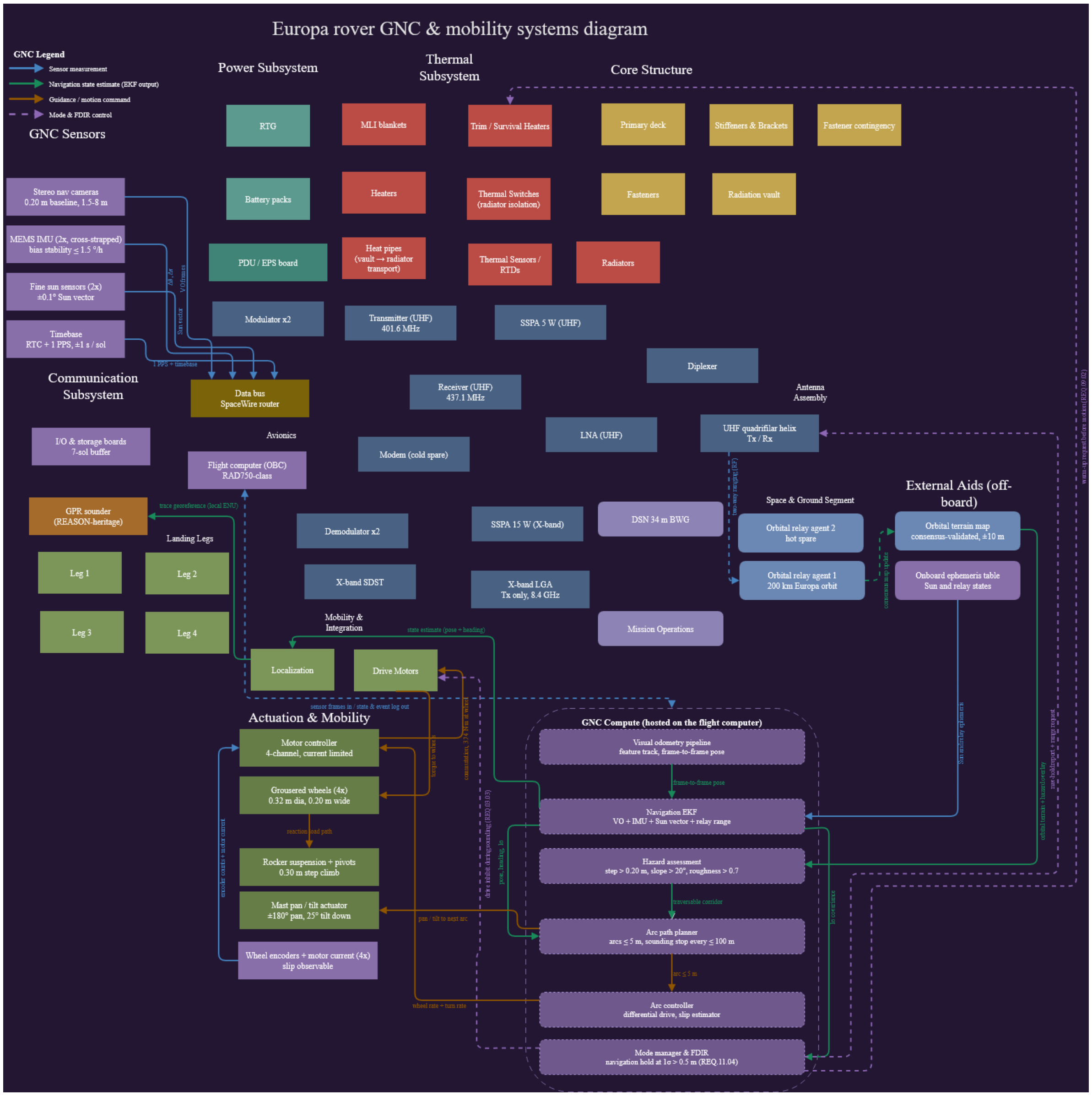



Figure 10: GNC and mobility systems diagram: sensor suite, GNC compute stack (visual odometry → navigation EKF → hazard assessment → arc planner → arc controller → mode manager/FDIR), and actuation/mobility hardware.

## 7.8 Command and Data Handling (CDH)

### 7.8.1 CDH Requirements

- **REQ.03.04**: Retain ≥ 7 sols of uncompressed radar trace data onboard; apply lossless compression at ≥ 2:1 to all Radar_Traces records before inter-agent relay transmission. (CDH, Science/Payload, COMMs)

- **REQ.04.01**: Support mesh-network consensus on shared terrain maps within ≤ 10 minutes under degraded (≤ 30% packet loss) link conditions. (CDH, COMMs)

- **REQ.05.02**: Detect and autonomously recover from a single-event latch-up in any processor core within 30 s of occurrence, restoring nominal operations without ground intervention, and log each event to non-volatile memory with a timestamp and affected core identifier. (CDH, Electronics)
- **REQ.12.01**: Provide $\geq$ 250 MIPS of radiation-hardened processing throughput with $\geq$ 30% processor duty-cycle and volatile-memory margin at the worst-case concurrent GNC compute load (visual odometry, hazard assessment, navigation EKF) within a single 5 m drive-arc cycle. (CDH)
- **REQ.12.02**: Provide $\geq$ 1 GB of EDAC-protected non-volatile mass storage sized to 7 sols of raw data at the 91.9 MB/sol generation rate with $\geq$ 30% margin, applying a science-priority retention policy that preserves NOMINAL-flagged radar traces and their paired localization records once remaining capacity falls below 10%. (CDH)
- **REQ.12.03**: Move all payload and navigation data on a cross-strapped SpaceWire primary bus $\geq$ 20 Mbps, with an independent RS-422 command/telemetry path $\geq$ 115.2 kbps that remains available with the primary bus failed. (CDH)
- **REQ.12.04**: Timestamp every record against a single onboard timebase held to $\pm$1 s absolute per sol and $\pm$10 ms record-to-record, framed with a trace identifier, sol index, and CCSDS-compliant packet header. (CDH)

### 7.8.2 CDH Architecture & Assumptions

The flight computer is a RAD750-class single-board computer, mounted inside the radiation vault (Section 7.3, REQ.07.03) alongside dedicated I/O and mass-storage boards, and it hosts the entire CDH function as well as the GNC compute stack described in Section 7.7. All payload and navigation data move on a cross-strapped SpaceWire router bus rated $\geq$ 20 Mbps: sustained rover-wide generation is only 6.3 kbps in idle modes, but a 1.57 MB stereo pair read out in about 2 s during Traverse is a 6.3 Mbps burst, and 20 Mbps clears that with a factor of 3.2 to spare (REQ.12.03). An independent RS-422 command and telemetry path ($\geq$ 115.2 kbps) carries safe-mode telemetry directly from the flight computer to the X-band SDST, bypassing the SpaceWire bus entirely, so a single bus failure cannot also take down the contingency beacon (REQ.10.03) that the comms architecture depends on as a relay-loss fallback.

Every record is timestamped against a single onboard timebase (RTC plus a 1 PPS distribution) held to $\pm$1 s absolute per sol and $\pm$10 ms record-to-record, and framed with a trace identifier, sol index, and CCSDS-compliant header (REQ.12.04): this is what lets the ground segment reassemble the Radar_Traces, Localization_Log, and Terrain_and_Health record sets by identifier alone after a lossy pass, rather than depending on packet arrival order. A memory scrubber runs a full EDAC scrub of mass memory on an hourly cycle, and the mode manager/FDIR function isolates a faulted data bus within $\leq$ 100 ms and drives safe-mode entry (REQ.08.03), detecting and recovering from a processor single-event latch-up within 30 s of occurrence (REQ.05.02). Figure 11 traces the full CDH core: SpaceWire interface, processing allocation, RS-422 backup, timebase/framing, EDAC/scrubbing, and mode manager/FDIR on one side; mass memory, the lossless compressor, OBC staging RAM, the downlink scheduler, science-priority retention, and the contingency-beacon fallback on the other, against every subsystem that feeds it data or draws on its outputs.

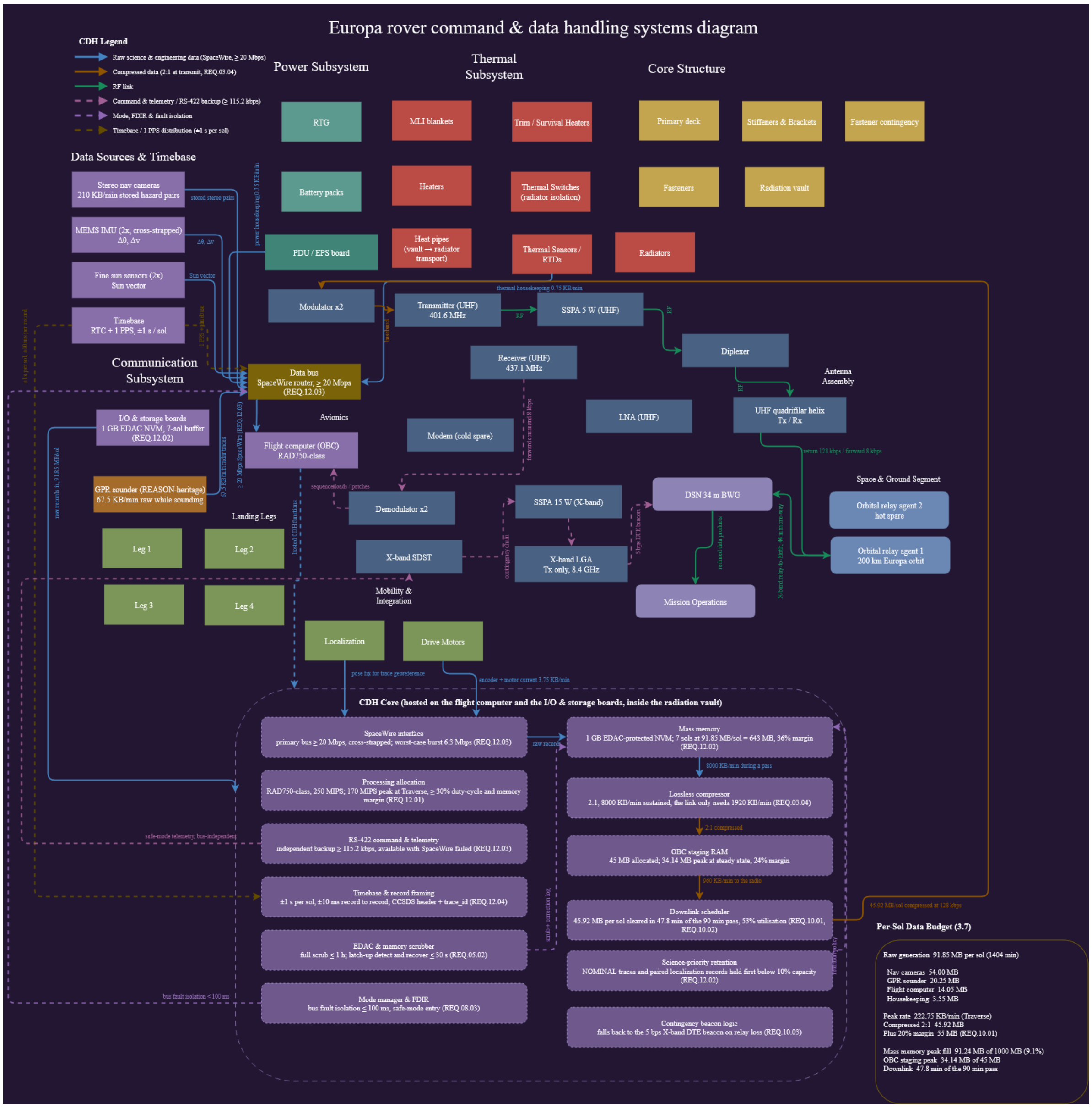



Figure 11: CDH systems diagram: data sources and timebase, the SpaceWire/RS-422 avionics core hosted on the RAD750-class flight computer, the mass-memory/compression/downlink pipeline, and the per-sol data budget (Section 7.7, REQ.03.04, REQ.05.02, REQ.10.01–REQ.10.03, REQ.12.01–REQ.12.04).

### 7.8.3 Data Budget and Storage Strategy

Table 24 breaks the 91.85 MB/sol raw-generation figure down by source. Nav-camera stereo hazard pairs dominate the budget even though they are the most compressible and most re-collectable product on board: the rover can re-image ground it drives again, but a discarded NOMINAL-flagged radar trace and its paired localization record cannot be recovered without re-driving the transect. That asymmetry is exactly why REQ.12.02's science-priority retention policy protects radar traces and their position fixes ahead of everything else once remaining mass-memory capacity drops below 10%.

| Source | Volume (MB/sol) |
|---|---|
| Nav cameras (stereo hazard pairs) | 54.00 |
| GPR sounder (radar traces) | 20.25 |
| Flight computer (state/sequencing) | 14.05 |
| Housekeeping (thermal/power/comms telemetry) | 3.55 |
| **Total raw generation** | **91.85** |

Table 24: Per-sol raw data generation by source.

Generation rate is far from uniform across modes (Table 25): Traverse is the binding case at 222.75 KB/min, driven by the nav-camera stereo pairs and the GNC compute allocation running together, while Night Low-Power and Battery Emergency fall to single digits. Radar Sounding and Traverse Planning tie at 76.5 KB/min because the sounder never runs concurrently with drive motors (REQ.03.03's stop-and-sound logic), so its 67.5 KB/min raw trace rate is the whole story in that mode, mirrored by the flight computer's own 30 KB/min map-generation-and-consensus compute load carried into Traverse Planning at 15 KB/min plus imagery.

| Operational Mode | Avg. Data Generation (KB/min) |
|---|---|
| Traverse | 222.75 |
| Radar Sounding | 76.5 |
| Traverse Planning | 76.5 |
| Map Gen & Consensus | 33.0 |
| Systems Warm-Up | 9.0 |
| Comms Relay Pass | 6.75 |
| Thermal Emergency | 5.25 |
| Night Low-Power | 3.0 |
| Battery Emergency | 2.25 |

Table 25: Average data-generation rate by operational mode.

Sized against a 1 GB EDAC-protected NAND storage board, 7 sols of raw data is 643 MB (REQ.03.04's storage horizon), which is 64% of device capacity and leaves 36% margin: the figure REQ.12.02 uses as its storage-sizing floor. In nominal operation, though, the buffer never gets close to that: the downlink scheduler clears each sol's compressed volume during the following relay pass, so mass memory peaks at only about 91.2 MB (9.1% of capacity) before the next contact drains it, and the full 643 MB horizon exists purely as an outage reserve if one or more relay passes are missed. The lossless compressor runs at 2:1 (REQ.03.04) with 8,000 KB/min of throughput margin against a sustained need of only about 1,920 KB/min, producing 45.92 MB of compressed data per sol; adding REQ.10.01's 20% margin gives the $\geq$ 55 MB/day delivered-volume floor that sizes the 128 kbps return link in Section 7.6. Compressed data queues in 45 MB of OBC staging RAM, which peaks at 34.14 MB (24% margin) at steady state: the tightest margin anywhere in the CDH design and the first place to look if the return link ever slows down or a relay outage lets a backlog build. The downlink scheduler clears the full 45.92 MB in 47.8 of the 90-minute relay pass (53% utilization, REQ.10.01/REQ.10.02), and if both relay links are lost the contingency-beacon logic falls back to the 5 bps X-band direct-to-Earth beacon from Section 7.6 (REQ.10.03) rather than waiting on the primary science-data path.

### 7.8.4 CDH Trade Study

**Study:** Analysis of the onboard flight computer and mass-storage architecture. Options were scored against five pairwise-weighted criteria (Mass 0.10, Power Demand 0.15, Processing Throughput 0.20, Radiation Durability 0.30, Reliability/TRL 0.25).

| Criterion (weight) | RAD750-class SBC + 1 GB rad-hard NAND | RAD5545 quad-core PowerPC + 4 GB | COTS ARM SoC (Zynq-class) + TMR voting |
|---|---|---|---|
| Mass (0.10) | 0.40 | 0.30 | 0.50 |
| Power Demand (0.15) | 0.60 | 0.45 | 0.75 |
| Processing Throughput (0.20) | 0.60 | 1.00 | 1.00 |
| Radiation Durability (0.30) | 1.50 | 1.20 | 0.60 |
| Reliability/TRL (0.25) | 1.25 | 0.75 | 0.50 |
| **Total Score** | **4.35** | 3.70 | 3.35 |

Table 26: Onboard flight computer and mass-storage architecture trade study.

**Winning option: RAD750-class SBC with a 1 GB rad-hard NAND storage board.** Radiation durability and flight heritage carry 55% of the weight between them, and that is what decides this. RAD750 parts are latch-up immune by process and qualified past 100 krad (the figure REQ.01.02 already fixes), and the part flew to Jupiter on Juno and is flying there again on Europa Clipper; nothing else in the trade has seen this environment. Throughput is where it gives ground: roughly 266 MIPS at 200 MHz against a peak load near 170 MIPS during Traverse, when visual odometry, hazard assessment, and the navigation EKF all run inside one arc cycle. That leaves 36% margin against the 30% floor in REQ.12.01, so the answer is adequate rather than comfortable, and it is the first term to watch if the visual-odometry pipeline grows. The RAD5545 buys throughput the mission does not need: peak generation across the whole rover is only 29.7 kbps and the buffer never climbs past 9% of capacity on a nominal sol, so there is no processing or storage problem worth its extra mass and lower TRL. The COTS ARM SoC is the tempting option and the wrong one; it wins mass, power, and throughput outright, then loses on the one axis that ends missions here: at 20–50 krad it is consumed inside the 30-sol surface phase even behind the vault, and TMR voting protects a running computation against upsets while doing nothing about total dose. With a 44-minute one-way light delay there is no ground recovery for a part that has simply worn out.

# 8 References

## References

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# Appendices

## A Extended Trade Study Tables

The Structures (Table 14), Power (Table 17), Thermal (Table 19), Communications (Table 21), GNC (Table 23), and CDH (Table 26) trade studies in Section 7 already reproduce the full pairwise-weighted decision matrices used in the source spreadsheets. Each uses the same methodology: criteria weighted by pairwise comparison in 0.25 increments (summing to 1 per pair, normalized to sum to unity across all criteria), and options scored 1–5 against a per-criterion rubric before weighting. Only the Structures and CDH scoring rubrics' full 1–5 band definitions are omitted here for length; they are reproduced in the source Decision Matrix / Trade Study PDFs in each subsystem's folder.

## B Full Requirements Spreadsheet

Table 27 consolidates every requirement identified across the source assignments (REQ.01.01 through REQ.12.04), with owning subsystem(s). Full justification, verification, and validation text for each requirement is maintained in the master spreadsheet (*documents/Space Robotics Mission Sheet.xlsx*, the cumulative requirements sheet, carried in this report's own folder) and is not fully duplicated here for length; representative full-text entries appear in Sections 3.3 and 7.

| ID | Statement (condensed) | Subsystem(s) |
|---|---|---|
| REQ.01.01 | Full autonomy for entire surface mission; no real-time Earth uplink. | System |
| REQ.01.02 | Survive $\geq 100$ krad TID over mission lifetime. | System, Electronics, Power |
| REQ.01.03 | $\geq 2$ active relay nodes at all times during surface ops. | System, COMMs |
| REQ.02.01 | Localization $\pm 0.5$ m over any 50 m segment, no GPS. | Mobility, GNC |
| REQ.03.01 | GPR resolves ice thickness $\pm 10\%$, 1–30 km depth, $\geq 100$ m resolution. | Science/Payload, Power |
| REQ.03.02 | Antenna boresight $\pm 5°$ nadir; halt above 20° slope. | Science/Payload, Structures, Mobility |
| REQ.03.03 | Peak power reserved for radar pulses; drive shutdown during sounding. | Science/Payload, Power, Mobility |
| REQ.03.04 | $\geq$ 7-sol onboard storage; $\geq$ 2:1 lossless compression before relay. | Science/Payload, COMMs, CDH |
| REQ.04.01 | Mesh consensus $\leq 10$ min under $\leq 30\%$ packet loss. | COMMs, CDH |
| REQ.05.01 | Avionics/battery 0–40°C active, $> -20$°C night mode. | Thermal, Power, Structures |
| REQ.05.02 | Detect/recover from a processor latch-up autonomously within 30 s, log event w/ timestamp & core ID. | CDH, Electronics |

| ID | Statement (condensed) | Subsystem(s) |
|---|---|---|
| REQ.05.03 | Per-sol draw $\leq$ 90% RTG output; $\geq$ 10% margin reserve. | Power, CDH, Thermal, Mobility |
| REQ.06.01 | $\geq$ 100 m/sol autonomous traverse; sounding stop $\leq$ 100 m intervals. | Mobility |
| REQ.06.02 | Onboard site-selection algorithm autonomously evaluates each NOMINAL-flagged radar trace against the cryobot insertion criteria (ice $<$ 5 km, slope $< 20°$, roughness $<$ 0.7) and flags a candidate site within one sol of the criteria first being satisfied, without ground intervention. | Science/Payload, Mobility, GNC |
| REQ.06.03 | Daily summary packet autonomously sent within 2 hr of sol's radar window. | COMMs, CDH, Science/Payload |
| REQ.07.01 | Landing legs accommodate $12g$ vertical / $6g$ lateral touchdown loads across four footpads, no permanent deformation. | Structures, Mobility, GNC |
| REQ.07.02 | Structural materials remain in-spec at 90 K over 30-sol lifetime. | Structures, Thermal, Power |
| REQ.07.03 | Radiation vault attenuates dose to $\leq$ 1 krad/sol at electronics level within structures mass allocation. | Structures, Power, Electronics, CDH |
| REQ.08.01 | MMRTG $\geq$ 110 We BOL, $\geq$ 80 We continuous, $\geq$ 1.9 kWh/sol. | Power, Thermal, System |
| REQ.08.02 | Battery $\geq$ 2.4 kWh usable; $\geq$ 12 hr night survival; $> -20°$C. | Power, Thermal |
| REQ.08.03 | PDU isolates a faulted load bus within 100 ms; critical loads uninterrupted; fault logged autonomously. | Power, CDH, Electronics |
| REQ.09.01 | Reject $\leq$ 500 W; vault $< 40°$C during peak ops. | Thermal, Electronics, Power, CDH |
| REQ.09.02 | Preheat actuators $> -40°$C within 60 min of cold start. | Thermal, Mobility, Structures |
| REQ.09.03 | MLI + RTG waste heat bound survival heaters $\leq$ 50 W. | Thermal, Power, Structures |
| REQ.10.01 | $\geq$ 55 MB/day compressed; $\geq$ 128 kbps return / $\geq$ 8 kbps forward. | COMMs, CDH, Science/Payload, Power |
| REQ.10.02 | $\geq$ 2 relay contacts/sol, $\geq$ 90 min total; latency $\leq$ 26 hr. | COMMs, CDH, Science/Payload, GNC |
| REQ.10.03 | Link $\geq$ 3 dB margin @ 588 km worst case; X-band DTE beacon $\geq$ 5 bps fallback. | COMMs, Power, Structures, CDH |
| REQ.11.01 | Attitude $\pm 0.5°$ roll/pitch, heading $\pm 0.25°$ per arc, sun-sensor fix every $\leq$ 5 m. | GNC, Mobility, Science/Payload, CDH |

| ID | Statement (condensed) | Subsystem(s) |
|---|---|---|
| REQ.11.02 | Hazard assessment $\geq$ 8 m lookahead; reject step $>$ 0.20 m / slope $>$ 20° / roughness $>$ 0.7. | GNC, Mobility, Science/Payload, CDH |
| REQ.11.03 | Cross-track $\leq$ 0.5 m/50 m; waypoint ±0.5 m; differential drive + slip estimate. | GNC, Mobility, Science/Payload |
| REQ.11.04 | Navigation hold if $1\sigma > 0.5$ m; re-init via 2-relay ranging + map correlation. | GNC, COMMs, CDH, Mobility |
| REQ.12.01 | $\geq$ 250 MIPS rad-hard throughput; $\geq$ 30% duty-cycle/memory margin at peak GNC compute load per arc. | CDH |
| REQ.12.02 | $\geq$ 1 GB EDAC NVM sized to 7 sols @ 91.9 MB/sol, $\geq$ 30% margin; science-priority retention below 10% capacity. | CDH |
| REQ.12.03 | SpaceWire primary bus $\geq$ 20 Mbps, cross-strapped; independent RS-422 backup $\geq$ 115.2 kbps survives primary-bus failure. | CDH |
| REQ.12.04 | Timebase ±1 s/sol, ±10 ms record-to-record; CCSDS header + trace ID framing. | CDH |

Table 27: Condensed master requirements list, REQ.01.01–REQ.12.04.

# C Sample Datasets

A 20-trace Radar_Traces export (Sol 1, `TR-0001`–`TR-0020`) and the complete Daily_Summary export (Sol 1–2, the full extent of that sheet's current contents), both pulled directly from *Data Product Space Robotics.xlsx*, are reproduced in Section 6.2 (Tables 7, 8, and 9) rather than duplicated here for length.

# D Tools Used

This report was assembled with the assistance of Claude (Anthropic), which synthesized and reorganized the author's prior assignment deliverables into the Section 4.1 template. Per-assignment AI-use disclosures carried over from the source documents are consolidated below.

- **2.1 Designing a Data Product:** "Artificial intelligence tools were used to verify that the technical statements, parameter values, and requirement references made in this document are consistent with the background research on Europa ice-shell radar sounding, cryobot mission concepts, and planetary robotics requirements engineering. AI also generated fake data to demonstrate the data being recorded."

- **2.3 ConOps Sketch:** "AI was used to check grammar and rephrase for better wording. The prompt used was: 'Please check grammar, cohesiveness and correct wording of the following."'

- **3.4 Point-Mass Thermal Equilibrium:** "AI was used to fact-check the results."

- **3.5 Link Budget:** "I used AI to check the arithmetic behind each link budget term and to cross-check the assumed parameters against published Electra and DSN performance data. I picked the architecture and the parameters myself, and I set the margin the same way."
- **3.6 Mobility Solution:** "I used AI to check the arithmetic behind the traction, torque, and localization error budget numbers, and to cross-check the low-temperature ice friction values against published measurements. The mobility architecture is all mine. Diagram generated using Claude Opus 5."
- **1.3 Deep Dive into a Robotic Mission:** "Claude Opus 5 (Anthropic) was used to compare this paper against the assignment rubric and provide a grade estimate."

**3.2 Structures and Mechanisms Design (dedicated AI Use Disclosure):** "The attached rover sketch was generated as an SVG graphic using Claude (model: Claude Opus 5), an AI assistant developed by Anthropic, accessed through the claude.ai interface. What the AI did: Produced the SVG drawing (linework, layout, labels, and leader lines) based on my direction. What the AI did not do: The design content, comprising component selection, structural configuration, materials (Al 7075-T73 struts, Ti-6Al-4V vault), payload (REASON-heritage GPR), and requirements traceability, comes from my own prior coursework (requirements, trade study, and mass budget for this mission). The AI drew from those documents rather than originating the design. Process: The sketch went through several revision cycles that I directed, including corrections to the suspension architecture, GPR mounting orientation (nadir-pointing, belly-mounted), sensor mast structure, RTG mounting, and the landing strut separation mechanism. I reviewed the final output for accuracy against my mission design."